

Care Commons EHR

ONC Health IT Certification
Software Manual

2015 Edition Cures Update

Care Commons Consortium

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Chapter 0

Installation, Runtime Environment, and Configuration

This chapter records the exact environment, dependencies, and configuration used to build the system and generate the certification test results in this manual. Version numbers are drawn from the repository as of the commit cited in the [Reproducibility](#) section; treat that commit plus `package-lock.json` as the authoritative build inputs.

0.1 System Requirements

Component	Requirement
Operating system	macOS (Apple Silicon or Intel) for development; Linux (Debian/Ubuntu) for CI and production; Windows via WSL2
Meteor	3.4 (<code>.meteor/release = METEOR@3.4</code>)
Node.js	22.22.0 — bundled by Meteor 3.4 as the app runtime; a system Node ≥ 20 (CI uses 20.11.0) for npm / CLI tooling
MongoDB	6.0 or newer — Meteor’s bundled development Mongo is 7.0.16; CI runs the <code>mongo:6.0</code> image
Browser (E2E)	Google Chrome / Chrome for Testing, with a version-matched ChromeDriver (the runs in this manual used 149.x)
Browser (app)	Any modern evergreen browser (Chrome, Edge, Firefox, Safari)
Desktop (opt.)	Electron via <code>npmPackages/desktop-lattice</code> — not required for certification testing

Table 1: System requirements

0.2 Package Inventory

Major dependencies, grouped by purpose. Versions are the declared ranges in `package.json`; exact resolved versions are pinned in `package-lock.json`.

Purpose	Package	Version
FHIR server	(in-repo) <code>server/FhirEndpoints.js</code> , <code>SearchParametersEngine</code>	—
	<code>@node-on-fhir/us-core</code> (US Core 7.0 profiles)	7.0.0
	<code>fhirpath</code>	4.8.0
SMART / Auth	<code>@node-oauth/express-oauth-server</code>	4.1.5
	<code>jsonwebtoken</code>	9.0.2
	<code>pkij</code> s + <code>asn1js</code> (X.509 / Backend Services)	3.2.4 / 3.0.5
	<code>bcrypt</code>	6.0.0
Data / Mongo	Meteor Mongo (server + Minimongo)	(Meteor 3.4)
	<code>simpl-schema</code> , <code>ajv</code> (FHIR JSON Schema)	— / 8.20.0
	<code>lodash</code> , <code>moment</code>	4.18.1 / 2.29.4
Testing	<code>nightwatch</code> (E2E)	3.12.1
	<code>chromedriver</code>	146.0.6

Purpose	Package	Version
	meteortesting:mocha (unit)	(Meteor)
UI	react / react-dom	18.2.0
	@mui/material / @mui/icons-material	5.16.7
C-CDA / QRDA	@node-on-fhir/ccda-export (C-CDA gen + receive)	(in-repo)
	@node-on-fhir/quality-measures (QRDA I export)	(in-repo)
	fqm-execution (FHIR measure engine)	1.8.5
	xml2js (XML parse / build)	0.6.2
Terminology	CodeSystems / ValueSets collections + US Core profiles	—
	code systems: CDCREC, LOINC, SNOMED CT, RxNorm (by reference)	—
DICOM	@cornerstonejs/core	1.78.0
	@cornerstonejs/dicom-image-loader / tools	1.78.0
Build tooling	Meteor 3.4 bundler	3.4
	@rspack/core (Rspack)	1.7.1
	tectonic (LaTeX — this manual only)	0.16.x

0.3 Installation

Meteor is the prerequisite; it provisions the bundled Node and MongoDB.

```
# 1. Install Meteor 3.x (once, system-wide)
curl https://install.meteor.com/ | sh

# 2. Clone and install
git clone https://github.com/node-on-fhir/core.git
cd core
npm install # installs npm workspaces + symlinks workflow packages

# 3. Run (starts the app + bundled MongoDB on http://localhost:3000)
npm start # equivalent to: meteor run
```

There is no separate `npm run dev` script; `npm start` maps to `meteor run`. A first run compiles the bundle and may take several minutes.

0.4 Environment Variables

The following variables configure the runtime; use safe placeholders in shared environments and never commit real secrets. Development / test values:

```
NODE_ENV=development # hard gate for dev auto-login
DEV_AUTO_LOGIN=true # enable dev auto-login
DEV_AUTO_USERNAME=demouser # dev account username
DEV_AUTO_PASSWORD=<dev-password> # dev account password (server-side only)
DEV_AUTO_PATIENT_ID=<fhir-id> # optional: bind dev user to a Patient
TEST_RUN=1 # gates destructive test-only operations
ENABLE_AUTOPUBLISH=true # enable autopublish.* publications
INITIALIZE_SEARCH_PARAMETERS=true # compile the FHIR SearchParametersEngine
```

```

# (REQUIRED for (g)(7)/(g)(9) search)
EXTRA_WORKFLOWS=@node-on-fhir/... # comma-separated workflow packages to load
MONGO_URL=mongodb://localhost:27017/meteor # override bundled dev Mongo
# (CI: mongodb://mongo:27017/meteor_test)
CHROMEDRIVER_PATH=/path/to/chromedriver # match your local Chrome version

```

Server configuration that is not an environment variable is supplied via a Meteor settings JSON file (`-settings`). Public keys land in `Meteor.settings.public` (client-visible); private keys stay server-side. Relevant fields (safe placeholders):

```

{
  "public": {
    "title": "Honeycomb",
    "fhirVersion": "R4",
    "devAutoLoginEnabled": true,
    "modules": {
      "patientDemographics": { "raceEthnicity": true }
    }
  },
  "private": {
    "fhir": {
      "rest": {
        "Patient": { "interactions": ["read","create","update","delete","search"],
          "search": true, "publication": true }
      },
      "disableOAuth": true
    },
    "accessControl": {
      "enableBasicAuth": false,
      "systemSecret": "<CHANGE-ME-do-not-commit>"
    }
  }
}

```

Security note. The certification/TDD settings file ships with `disableOAuth: true` so unauthenticated REST calls are granted a `noauth` role — convenient for a self-contained test run, but never a production posture. Production deployments must enable OAuth / SMART, set a real `systemSecret` (or migrate to JWT/JWK Backend Services), and keep settings files containing secrets out of version control.

0.5 Canonical Certification Configuration

The certification test run in this manual was launched with the following exact command — environment variables, the workflow-package set, and the settings file. This is the reproducible entry point for the whole suite:

```

TEST_RUN=1 \
ENABLE_AUTO PUBLISH=true \
DEV_AUTO_LOGIN=true \
DEV_AUTO_USERNAME=demouser \
DEV_AUTO_PASSWORD=<dev-password> \
INITIALIZE_SEARCH_PARAMETERS=true \

```



```
EXTRA_WORKFLOWS=@node-on-fhir/order-catalog,@node-on-fhir/implantable-devices,@node-on-
  fhir/ccda-export,@node-on-fhir/prescription-benefit,@node-on-fhir/decision-support,
  @node-on-fhir/quality-measures,@node-on-fhir/secure-messaging \
meteor run --settings settings/settings.honeycomb.tdd.json
```

The settings file `settings/settings.honeycomb.tdd.json` enables dev auto-login, the per-resource FHIR REST interactions (`private.fhir.rest.*`), and the race/ethnicity demographics gate (`public.modules.patientDemographics.raceEthnicity = true`, required for the (a)(5) test). Meteor reads `-settings` into `Meteor.settings` at boot. The seven `EXTRA_WORKFLOWS` packages supply the CPOE catalog, implantable-device registry, C-CDA export/receive, real-time prescription benefit, decision support, quality measures, and secure messaging surfaces exercised by the criteria chapters.

0.6 Test Execution

Run the full Base EHR suite against a running server with the aggregate runner:

```
./certification/tdd/base_ehr/run-base-ehr-tests.sh
```

Or run a single criterion directly (the form used to capture the per-chapter results and screenshots in this manual):

```
CHROMEDRIVER_PATH=/path/to/chromedriver \
npx nightwatch --config nightwatch.circle.conf.js \
certification/tdd/base_ehr/170.315.<x>.<y>.test.js
```

Each behavioral test drives real record / change / access behaviors against FHIR resources per the ONC test procedures; a criterion is **Verified** when its test passes green and a **Gap** when a test deliberately fails on a documented missing capability. See the Certification Testing Notes appendix for the full suite status and the relationship to Inferno.

0.7 Reproducibility

The results, screenshots, and status table in this manual were generated with:

Input	Value
Repository	github.com/node-on-fhir/core (branch <code>baseehr-testcoverage</code>)
Commit SHA	c4ad354131cea9801a585b76ff04a629939eab24 (documentation-only changes may follow this commit)
package-lock.json	lockfileVersion 3; SHA-256 a1a97949761c95fcde77784cc2a4c647ed9748443744c31965f7d3fd4a73782b (4,122 locked packages)
Node.js	22.22.0 (Meteor 3.4 runtime); 20.11.0 (CI tooling)
MongoDB	7.0.16 (bundled dev); 6.0 (CI)
Meteor	3.4
Browser	Chrome for Testing with version-matched ChromeDriver (149.x)

Table 3: Reproducibility inputs

To reproduce, check out the commit above, restore dependencies from `package-lock.json` with `npm ci`, launch with the [Canonical Certification Configuration](#), and run the test-execution commands above.

Part I

Base EHR Certification

Overview

This Part documents the Base EHR certification criteria per the ONC Health IT Certification Program, updated for the **CY2026 Base EHR definition** (45 CFR 170.102, as revised by the HTI final rules). The CPOE family (a)(1)–(a)(3) requires at least one member; the remaining rows are each required (some with future compliance dates, noted below).

Certification Criteria Summary

The **Status** column reflects the outcome of the behavioral Nightwatch test suite under `certification/tdd/base_e` (one test file per criterion). **Verified** = the capability is built and its behavioral tests pass green. **Gap** = a behavioral test exists and deliberately fails on a documented missing capability (the suite is a live certification punch-list — see the per-criterion chapters and the Certification Testing Notes appendix). **Inferno** = certified externally by the ONC-approved Inferno tool, not by this suite.

Criterion	Title	Status
§ 170.315(a)(1)	CPOE - Medications	Verified
§ 170.315(a)(2)	CPOE - Laboratory	Verified
§ 170.315(a)(3)	CPOE - Diagnostic Imaging	Verified
§ 170.315(a)(5)	Demographics	Verified
§ 170.315(a)(14)	Implantable Device List	Verified
§ 170.315(b)(1)	Transitions of Care (C-CDA)	Verified
§ 170.315(b)(4)	Real-Time Prescription Benefit	Verified
§ 170.315(b)(11)	Decision Support Interventions	Verified
§ 170.315(c)(1)	Clinical Quality Measures - Record & Export	Verified
§ 170.315(g)(7)	Application Access - Patient Selection	Verified
§ 170.315(g)(9)	Application Access - All Data Request	Verified
§ 170.315(g)(10)	Standardized API	Inferno
§ 170.315(h)(1)	Direct Project (or (h)(2))	Gap

Table 4: CY2026 Base EHR Certification Criteria — test-verified status (11 verified, 1 documented gap, 1 external/Inferno)

The remaining documented gap is (h)(1) Direct messaging, which has no operational transport (SMTP/S-MIME/HISP) and uses simulated certificate trust; it is detailed in its chapter and tracked in the project’s engineering ledger. The former (a)(5), (b)(1), and (c)(1) gaps have since been remediated: (a)(5) now records race/ethnicity per CDCREC as US Core extensions (settings-gated — see that chapter); (b)(1) now populates the C-CDA from the real patient record and adds an inbound receive/validate/display path; and (c)(1) now exports a QRDA Category I document

per § 170.205(h)(2).

Note on future compliance dates: (b)(4) Real-Time Prescription Benefit is an HTI-4 criterion required by 1/1/2028, and (b)(11) Decision Support Interventions has a Privacy & Security update due 12/31/2027. Both are tested at their current baseline; absent future-dated portions are treated as informational, not gaps.

Legacy / non-core criteria documented separately: § 170.315(a)(4) Drug Interaction Checks, § 170.315(a)(9) Clinical Decision Support (**expired** 1/1/2025, replaced by (b)(11)), and § 170.315(e)(1) View, Download, Transmit are retained as reference chapters below but are not part of the CY2026 Base EHR core set above.

Chapter 1

§ 170.315(a)(1) - CPOE Medications

1.1 Criterion Information

1.1.1 Maturity Level

VERIFIED (behavioral tests green)

1.1.2 Regulatory Text

From 45 CFR § 170.315(a)(1):

- (1) **Computerized provider order entry—medications.**
 - (i) Enable a user to record, change, and access medication orders.
 - (ii) Optional. Include a “reason for order” field.

1.2 Behavior-Driven Development Specification

Listing 1.1: 170.315(a)(1) - CPOE Medications

```
# certification/bdd/170.315-a-1-cpoe-medications.feature
# §170.315(a)(1) - Computerized Provider Order Entry - Medications

# REGULATORY TEXT FROM 45 CFR §170.315(a)(1)
#
# (1) Computerized provider order entrymedications.
#
# (i) Enable a user to record, change, and access medication orders.
#
# (ii) Optional. Include a "reason for order" field.

@170.315-a-1 @cpoe @medications @clinical
Feature: Computerized Provider Order Entry - Medications
  As a provider
  I want to create and manage medication orders

  Background:
    Given I am authenticated as a provider
    And a patient record is selected
    And I am on /order-catalog

  Scenario: Record a new medication order
    Given the patient requires "Metformin_500_MG_Oral_Tablet"
    When I open the OrderCatalog and select the medications order type
    And select the medication from the catalog
    And provide an optional "reason_for_order" of "Type_2_diabetes_mellitus"
    And submit the order
    Then the system shall create a FHIR MedicationRequest with status "active" and intent
         "order"
    And the MedicationRequest shall carry the RxNorm-coded medication, requester, and
         dosage instructions
    And the MedicationRequest shall be inserted into the MedicationRequests collection
    And an AuditEvent shall record the order creation
```

Scenario: Change an existing medication order

Given the patient has a pending MedicationRequest

When I change the MedicationRequest status to "on-hold" and priority to "urgent"

Then the modified values shall be persisted in the MedicationRequests collection

And the change shall be retrievable via the medication orders API

Scenario: Access existing medication orders

Given the patient has one or more MedicationRequests

When I navigate to /medication-requests

Then the system shall update the MedicationRequests subscription by patientId

And the MedicationRequestsTable shall display the patient's medication orders

And I shall be able to open a MedicationRequest to view its details

1.3 Screenshots

Honeycomb CLEAR PATIENT demouser LOGOUT

BaseEHR CpoeMeds ID: baseehr-a1-1783151151392 Birth Date: May 05, 1985 Gender: other Phone: Unknown

Computerized Provider Order Entry (CPOE)
ONC §170.315(a)(1) Medications | §170.315(a)(2) Laboratory | §170.315(a)(3) Diagnostic Imaging

Order Catalog
Browse medications

LABORATORY **MEDICATION** RADIOLOGY

Search medication orders...

All Categories Antibiotic Analgesic NSAID Opioid Beta Blocker ACE Inhibitor Diuretic Antidiabetic

Code	Description	Route	Strength	Action
308182	Amoxicillin 500 MG Oral Capsule Antibiotic	PO	500 mg	+
308459	Azithromycin 250 MG Oral Tablet Antibiotic	PO	250 mg	+
308745	Ceftriaxone 1 GM Injection Antibiotic	IM/IV	1 g	+
313782	Acetaminophen 325 MG Oral Tablet Analgesic	PO	325 mg	+

Active Orders
1 items pending submission

Default Priority
☒ Routine ☐ Urgent ☐ STAT

routine Metformin 500 MG Oral Tablet

CLEAR ALL SUBMIT ORDERS (1)

ONC Requirement: All orders must include priority level and clinical indication when applicable.

Figure 1.1: CPOE Medications - Order Entry Interface

Chapter 2

§ 170.315(a)(2) - CPOE Laboratory

2.1 Criterion Information

2.1.1 Maturity Level

VERIFIED (behavioral tests green)

2.1.2 Regulatory Text

From 45 CFR § 170.315(a)(2):

- (2) **Computerized provider order entry—laboratory.**
 - (i) Enable a user to record, change, and access laboratory orders.
 - (ii) Optional. Include a “reason for order” field.

2.2 Behavior-Driven Development Specification

Listing 2.1: 170.315(a)(2) - CPOE Laboratory

```
# /Volumes/SonicMagic/Code/honeycomb-public-release/certification/bdd/170.315-a-2-cpoe-
  laboratory.feature
# §170.315(a)(2) - Computerized Provider Order Entry - Laboratory

# REGULATORY TEXT FROM 45 CFR §170.315(a)(2)
#
# (2) Computerized provider order entrylaboratory.
#
# (i) Enable a user to record, change, and access laboratory orders.
#
# (ii) Optional. Include a "reason for order" field.

@170.315-a-2 @cpoe @laboratory @clinical
Feature: Computerized Provider Order Entry - Laboratory
  As a provider
  I want to create and manage laboratory orders

Background:
  Given I am authenticated as a provider
  And a patient record is selected
  And I am on /order-catalog/laboratory

Scenario: Create a new laboratory order
  Given the patient requires a "Complete_Blood_Count"
  When I open the OrderCatalog on /order-catalog
  And select a PlanDefinition for "CBC_with_differential"
  Then the system shall reference the PlanDefinition.id of "CBC_with_differential"
  And look up the associated actions and ActivityDefinitions
  And create a new resource based on ActivityDefinition.kind (typically a
    ServiceRequest)
  And the ServiceRequest shall include all necessary details for specimen collection
  And the ServiceRequest shall be inserted into the ServiceRequests collection

Scenario: Change an existing laboratory order
```

Given the patient has a pending ServiceRequest for "Basic_Metabolic_Panel"
 When I change the ServiceRequest to "Comprehensive_Metabolic_Panel"
 And the modified ServiceRequest shall be saved in the ServiceRequests collection

Scenario: Access existing laboratory orders

Given the patient has multiple ServiceRequests in various states
 When I request to view the patient's ServiceRequests
 Then the system shall update the ServiceRequests subscription by patientId
 And the ServiceRequestsTable shall show current status (pending, completed, cancelled) for each request
 And I shall be able to view results for completed orders

Background:

Given I am authenticated as a lab tech

Scenario: Access workqueue

Given no patient record is selected
 And I am on /service-requests
 Then the system shall enable access to all ServiceRequests of type "laboratory"
 And the orders shall show current status
 And I shall be able to open the ServiceRequest records
 And update their status and other fields

2.3 Screenshots

The screenshot displays the Honeycomb BaseEHR CPOeLab interface. The header shows patient information: ID baseehr-a2-1783151186638, Birth Date May 05, 1985, Gender other, and Phone Unknown. The main content area is titled "Computerized Provider Order Entry (CPOE)" and includes a "VIEW SERVICE REQUESTS" button. The "Order Catalog" section allows browsing laboratory tests, with a search bar and filters for Chemistry, Hematology, Coagulation, Microbiology, and Urinalysis. A table lists laboratory orders with columns for Code, Description, Specimen, TAT, and Action. The "Active Orders" section shows 1 item pending submission, with a default priority of Routine and a "SUBMIT ORDERS (1)" button. A notification bar at the bottom states: "ONC Requirement: All orders must include priority level and clinical indication when applicable."

Code	Description	Specimen	TAT	Action
2823-3	Potassium [Moles/volume] in Serum or Plasma Chemistry	Serum/Plasma	4 hours	+
2345-7	Glucose [Mass/volume] in Serum or Plasma Chemistry	Serum/Plasma	4 hours	+
2160-0	Creatinine [Mass/volume] in Serum or Plasma Chemistry	Serum/Plasma	4 hours	+

Figure 2.1: CPOE Laboratory - Order Entry Interface

Chapter 3

§ 170.315(a)(3) - CPOE Diagnostic Imaging

3.1 Criterion Information

3.1.1 Maturity Level

VERIFIED (behavioral tests green)

3.1.2 Regulatory Text

From 45 CFR § 170.315(a)(3):

- (3) Computerized provider order entry—diagnostic imaging.**
 - (i) Enable a user to record, change, and access diagnostic imaging orders.
 - (ii) Optional. Include a “reason for order” field.

3.2 Behavior-Driven Development Specification

Listing 3.1: 170.315(a)(3) - CPOE Diagnostic Imaging

```
# /Volumes/SonicMagic/Code/honeycomb-public-release/certification/bdd/170.315-a-3-cpoe-
  diagnostic-imaging.feature
# §170.315(a)(3) - Computerized Provider Order Entry - Diagnostic Imaging

# REGULATORY TEXT FROM 45 CFR §170.315(a)(3)
#
# (3) Computerized provider order entry diagnostic imaging.
#
# (i) Enable a user to record, change, and access diagnostic imaging orders.
#
# (ii) Optional. Include a "reason for order" field.

@170.315-a-3 @cpoe @diagnostic-imaging @clinical
Feature: Computerized Provider Order Entry - Diagnostic Imaging
  As an ordering provider
  I want to electronically create and manage diagnostic imaging orders
  So that I can order appropriate imaging studies efficiently

Background:
  Given I am authenticated as a physician
  And a patient record is selected
  And I am on /order-catalog/diagnostic-imaging

Scenario: Create a new diagnostic imaging order
  Given the patient requires a chest X-ray
  When I search the PlanDefinitions collection for "Chest_X-ray_PA_and_lateral"
  Then the system shall open a new ServiceRequest
  And the ServiceRequest shall include imaging modality and anatomical location
  And the new record will be inserted into the ServiceRequests collection

Scenario: Change an existing diagnostic imaging order
  Given the patient has a pending ServiceRequest for "Chest_X-ray_PA_and_lateral"
  When I open the ServiceRequest and change it to "CT_abdomen_and_pelvis_with_contrast"
  and save
```

Then the system shall update the record in the ServiceRequests collection
And it will be viewable at /service-requests

Scenario: Optionally include reason for diagnostic imaging order
Given I am creating a new ServiceRequest for "MRI_lumbar_spine"
Then the system may include a "reason_for_order" field

Scenario: Access existing diagnostic imaging
Given a patient record is selected
And I am on /imaging-studies
And the patient has multiple imaging orders and completed studies (CT abdomen pelvis;
MRI lumbar)
Then the system shall enable access to all ImagingStudies for the selected patient
And the orders shall show current status
And I shall be able to access imaging reports and images (DocumentReferences)

Background:
Given I am authenticated as a rad tech

Scenario: Access workqueue
Given no patient record is selected
And I am on /service-requests
Then the system shall enable access to all ServiceRequests of type "imaging"
And the orders shall show current status
And I shall be able to open the ServiceRequest records
And update their status and other fields

Scenario: Access existing diagnostic imaging
Given a patient record is selected
And I am on /imaging-studies
And the patient has multiple imaging orders and completed studies (CT abdomen pelvis;
MRI lumbar)
Then the system shall enable access to all ImagingStudies for the selected patient
And the orders shall show current status
And I shall be able to access imaging reports and images (DocumentReferences)

3.3 Screenshots

CLEAR PATIENT
demouser
LOGOUT

BaseEHR Cpoelming
ID: baseehr-a3-1783151215465
Birth Date: Jan 01, 1970
Gender: female
Phone: Unknown

Computerized Provider Order Entry (CPOE)
[VIEW SERVICE REQUESTS](#)

Order Catalog
Browse radiology procedures

LABORATORY
MEDICATION
RADIOLOGY

All Categories
X-Ray
CT
MRI
Ultrasound
Nuclear Medicine
Mammography
Fluoroscopy
Angiography

Code	Description	Modality	Body Part	TAT	Action
36643-5	XR Chest 1 View X-Ray	XR	Chest	2 hours	+
30746-2	XR Chest 2 Views X-Ray	XR	Chest	2 hours	+
24627-2	XR Chest PA and Lateral X-Ray	XR	Chest	2 hours	+
29252-4	CT Chest without contrast CT	CT	Chest	4 hours	+

Active Orders
1 items pending submission

Default Priority
☒ Routine
☐ Urgent
☐ STAT

routine
XR Chest 2 Views

CLEAR ALL
SUBMIT ORDERS (1)

ONC Requirement: All orders must include priority level and clinical indication when applicable.

Figure 3.1: CPOE Diagnostic Imaging - Order Entry Interface

Chapter 4

§ 170.315(a)(4) - Drug Interaction Checks

4.1 Criterion Information

4.1.1 Maturity Level

IMPLEMENTED

4.1.2 Regulatory Text

From 45 CFR § 170.315(a)(4):

(4) Drug-drug, drug-allergy interaction checks for CPOE —

(i) Interventions. Before a medication order is completed and acted upon during computerized provider order entry (CPOE), interventions must automatically indicate to a user drug-drug and drug-allergy contraindications based on a patient's medication list and medication allergy list.

(ii) Adjustments.

(A) Enable the severity level of interventions provided for drug-drug interaction checks to be adjusted.

(B) Limit the ability to adjust severity levels in at least one of these two ways:

(1) To a specific set of identified users.

(2) As a system administrative function.

4.2 Behavior-Driven Development Specification

Listing 4.1: 170.315(a)(4) - Drug Interaction Checks

```
# /Volumes/SonicMagic/Code/honeycomb-public-release/certification/bdd/170.315-a-4-drug-
  interactions.feature
# §170.315(a)(4) - Drug-Drug, Drug-Allergy Interaction Checks for CPOE

# REGULATORY TEXT FROM 45 CFR §170.315(a)(4)
#
# (4) Drug-drug, drug-allergy interaction checks for CPOE
#
# (i) Interventions. Before a medication order is completed and acted upon during
  computerized provider order entry (CPOE), interventions must automatically indicate
  to a user drug-drug and drug-allergy contraindications based on a patient's
  medication list and medication allergy list.
#
# (ii) Adjustments.
#
# (A) Enable the severity level of interventions provided for drug-drug interaction
  checks to be adjusted.
#
# (B) Limit the ability to adjust severity levels in at least one of these two ways:
#
# (1) To a specific set of identified users.
#
# (2) As a system administrative function.

@170.315-a-4 @cpoe @drug-interactions @safety @clinical
```

Feature: Drug-Drug and Drug-Allergy Interaction Checks for CPOE

As a prescribing provider

I want automatic alerts for potential drug interactions

So that I can prevent adverse drug events and improve patient safety

Background:

Given I am authenticated as a provider

And a patient record is selected

And the patient has an active medication list

And the patient has a medication allergy list

And I am on /order-catalog/medications

Scenario: Automatic drug-drug interaction checking during CPOE

Given the patient is currently taking "Warfarin_5mg_daily"

When I am on /order-catalog/medications

Then an RxNorm search bar will display, with RxNorm accordion table underneath with top-10 records

When I select the medication request for "Aspirin_325mg_daily"

Then the row will expand, with action buttons

And the system shall automatically indicate drug-drug contraindications with an alert

And the alert shall describe the interaction between Warfarin and Aspirin

Scenario: Automatic drug-allergy interaction checking during CPOE

Given the patient has a documented allergy to "Penicillin"

When I am on /order-catalog/medications

Then an RxNorm search bar will display, with RxNorm accordion table underneath with top-10 records

When I complete the medication request for "Amoxicillin_500mg"

Then the row will expand, with action buttons

Then the system shall automatically indicate drug-allergy contraindications with an alert

And the alert shall identify the cross-reactivity concern

Scenario: Display interventions based on active medication list

Given the patient's active medication list contains multiple medications

When I search for medications

Then the system shall check against all medications in the active medication list

And all drug-drug interactions shall be displayed as chips or icons in the accordion table

Scenario: Display interventions based on medication allergy list

Given the patient's allergy list contains documented allergies

When I search for medications

Then the system shall check against all allergies in the medication allergy list

And all relevant drug-allergy contraindications shall be displayed as chips or icons in the accordion table

Scenario: Enable severity level adjustment for drug-drug interactions

Given I am a system administrator

When I access the /drug-drug-interaction-settings

Then the system shall enable adjustment of severity levels

And severity levels shall control which interactions trigger alerts

Scenario: Display high-severity drug-drug interaction

Given the interaction checking system is configured with severity levels

And the patient is taking "MAO_oInhibitor"

When I attempt to order "SSRI_oantidepressant"

Then the system shall display high-severity interaction alert

And the alert shall require acknowledgment

Scenario: Display moderate-severity drug-drug interaction

Given the interaction checking system is configured with severity levels

And the patient is taking "Atorvastatin"

When I attempt to order "Clarithromycin"

Then the system shall display moderate-severity interaction alert

And the alert shall provide clinical guidance

Scenario: Provider override of drug interaction alert

Given a drug-drug interaction alert is displayed

When I review the alert and determine the benefit outweighs risk

Then the system shall allow me to 'override' the alert

And the override shall expand a collapsed clinical reasoning input

And the override shall be recorded in the audit log

Scenario: Multiple simultaneous interaction alerts

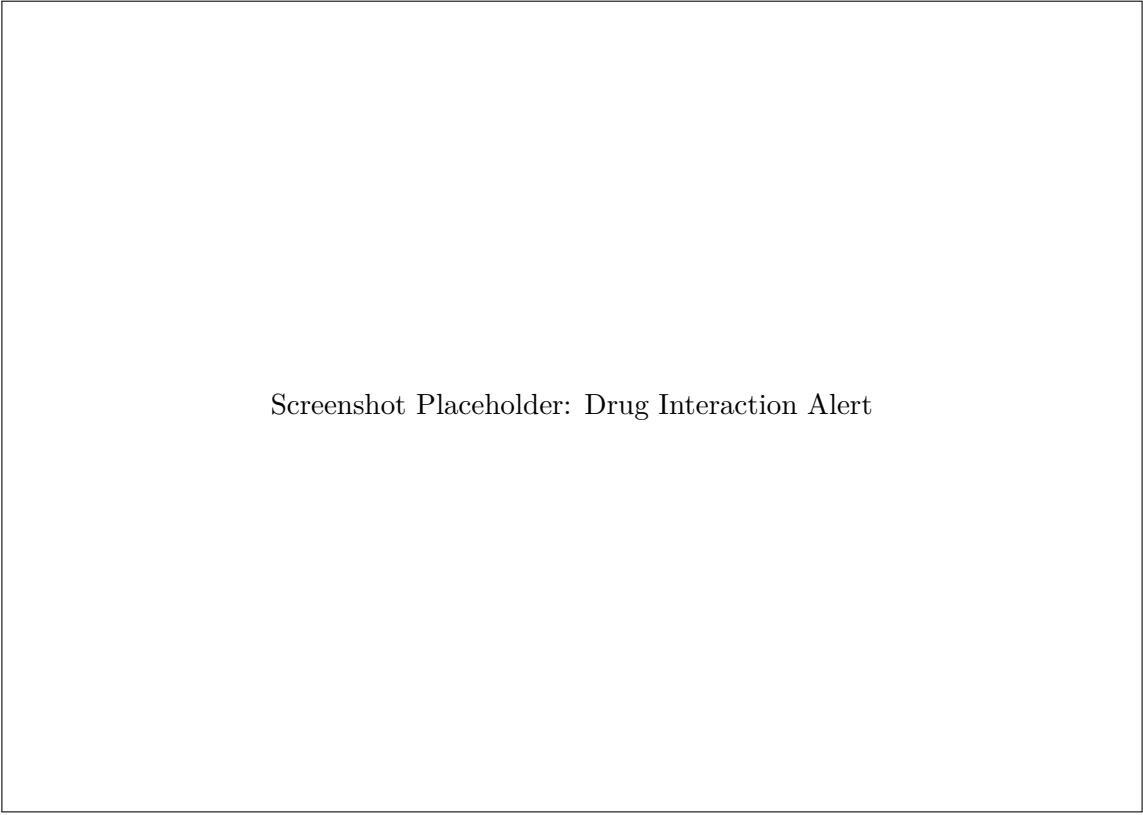
Given the patient has complex medication regimen

When I select a medication that interacts with multiple existing medications or allergies

Then the system shall display all relevant interactions in the expansion panel

And interactions shall be prioritized by severity

4.3 Screenshots



Screenshot Placeholder: Drug Interaction Alert

Figure 4.1: Drug-Drug Interaction Alert Display

Chapter 5

§ 170.315(a)(5) - Demographics

5.1 Criterion Information

5.1.1 Maturity Level

VERIFIED (behavioral tests green)

5.1.2 Regulatory Text

From *45 CFR § 170.315(a)(5)*:

(5) Patient demographics and observations.

(i) Enable a user to record, change, and access patient demographic and observations data including race, ethnicity, preferred language, sex, sex parameter for clinical use, sexual orientation, gender identity, name to use, pronouns, and date of birth.

(Full regulatory text abbreviated for brevity. See BDD specification for complete requirements.)

5.1.3 Implementation

The behavioral test records, changes, and accesses name, date of birth, sex (US Core birth-sex, with a decline-to-specify option), and preferred language (RFC 5646), and now **race and ethnicity** per the CDCREC code system aggregated to OMB categories, with a declined-to-specify option. The patient form provides multi-select Race and Ethnicity fields, and the patient-write methods preserve the complex (nested sub-extension) US Core `us-core-race` / `us-core-ethnicity` extensions.

Settings gate (international deployments): collecting race/ethnicity is legally forbidden in some jurisdictions, so it is opt-in — the fields render, and the extensions persist, only when `settings.public.modules.patientDemographics.raceEthnicity` is enabled; when disabled the fields are hidden and the server strips the extensions defensively. US certification builds enable the flag.

Regulatory note (CCG v1.3, updated 05/2026): consistent with Executive Order 14168 and OPM guidance, sexual orientation, gender identity, sex parameter for clinical use, name to use, and pronouns are *not* required for (a)(5); the sex requirement reduces to recording Female or Male (plus decline).

5.2 Behavior-Driven Development Specification

Listing 5.1: 170.315(a)(5) - Demographics

```
# /Volumes/SonicMagic/Code/honeycomb-public-release/certification/bdd/170.315-a-5-
  demographics-observations.feature
# §170.315(a)(5) - Patient Demographics and Observations

# REGULATORY TEXT FROM 45 CFR §170.315(a)(5)
#
# (5) Patient demographics and observations.
#
# (i) Enable a user to record, change, and access patient demographic and observations
  data including race, ethnicity, preferred language, sex, sex parameter for clinical
  use, sexual orientation, gender identity, name to use, pronouns, and date of birth.
#
```

(A) Race and ethnicity.

#

(1) Enable each one of a patient's races to be recorded in accordance with, at a minimum, the standard specified in §170.207(f)(3) and whether a patient declines to specify race.

#

(2) Enable each one of a patient's ethnicities to be recorded in accordance with, at a minimum, the standard specified in §170.207(f)(3) and whether a patient declines to specify ethnicity.

#

(3) Aggregate each one of the patient's races and ethnicities recorded in accordance with paragraphs (a)(5)(i)(A)(1) and (2) of this section to the categories in the standard specified in §170.207(f)(1).

#

(B) Preferred language. Enable preferred language to be recorded in accordance with the standard specified in §170.207(g)(2) and whether a patient declines to specify a preferred language.

#

(C) Sex. Enable sex to be recorded in accordance with the standard specified in § 170.207(n)(1) for the period up to and including December 31, 2025; or §170.207(n)(2).

#

(D) Sexual orientation. Enable sexual orientation to be recorded in accordance with, at a minimum, the version of the standard specified in §170.207(o)(1) for the period up to and including December 31, 2025; or §170.207(o)(3), as well as whether a patient declines to specify sexual orientation.

#

(E) Gender identity. Enable gender identity to be recorded in accordance with, at a minimum, the version of the standard specified in §170.207(o)(2) for the period up to and including December 31, 2025; or §170.207(o)(3), as well as whether a patient declines to specify gender identity.

#

(F) Sex Parameter for Clinical Use. Enable at least one sex parameter for clinical use to be recorded in accordance with, at a minimum, the version of the standard specified in §170.207(n)(3). Conformance with this paragraph is required by January 1, 2026.

#

(G) Name to Use. Enable at least one preferred name to use to be recorded. Conformance with this paragraph is required by January 1, 2026.

#

(H) Pronouns. Enable at least one pronoun to be recorded in accordance with, at a minimum, the version of the standard specified in §170.207(o)(4). Conformance with this paragraph is required by January 1, 2026.

#

(ii) Inpatient setting only. Enable a user to record, change, and access the preliminary cause of death and date of death in the event of mortality.

@170.315-a-5 @demographics @patient-data @clinical

Feature: Patient Demographics and Observations

As a healthcare provider

I want to record comprehensive patient demographic and observation data

So that I can provide culturally competent and personalized care

Background:

Given I am authenticated as a provider
And a patient record is selected

----- *Race and Ethnicity* -----

Scenario: Record patient's race using standard vocabulary

Given I am recording patient demographics
When I record the patient's race
Then the system shall enable recording per §170.207(f)(3) standard
And the system shall support granular race categories
And the system shall enable recording multiple races

Scenario: Record multiple races for a patient

Given a patient identifies with multiple racial categories
When I record the patient's races
Then the system shall enable each one of the patient's races to be recorded
And each race shall be stored as discrete data element
And all recorded races shall be retrievable

Scenario: Record patient decline to specify race

Given I am recording patient demographics
When the patient declines to specify their race
Then the system shall enable recording that patient declines to specify race
And this shall be stored as distinct from missing data

Scenario: Record patient's ethnicity using standard vocabulary

Given I am recording patient demographics
When I record the patient's ethnicity
Then the system shall enable recording per §170.207(f)(3) standard
And the system shall support Hispanic/Latino ethnicity categories

Scenario: Record multiple ethnicities for a patient

Given a patient identifies with multiple ethnic categories
When I record the patient's ethnicities
Then the system shall enable each one of the patient's ethnicities to be recorded
And each ethnicity shall be stored as discrete data element

Scenario: Record patient decline to specify ethnicity

Given I am recording patient demographics
When the patient declines to specify their ethnicity
Then the system shall enable recording that patient declines to specify ethnicity
And this shall be stored as distinct from missing data

Scenario: Aggregate race and ethnicity to standard categories

Given granular race and ethnicity data has been recorded
When aggregating for reporting purposes
Then the system shall aggregate to categories per §170.207(f)(1)
And aggregation shall maintain data integrity
And original granular data shall be preserved

----- *Preferred Language* -----

Scenario: Record patient's preferred language

Given I am recording patient demographics

When I record the patient's preferred language
 Then the system shall enable recording per §170.207(g)(2) standard
 And the system shall use ISO 639-2 language codes

Scenario: Record patient decline to specify preferred language
 Given I am recording patient demographics
 When the patient declines to specify preferred language
 Then the system shall enable recording that patient declines to specify
 And this shall be stored as distinct from missing data

----- Sex -----

Scenario: Record patient's sex (until December 31, 2025)
 Given the current date is before December 31, 2025
 When I record the patient's sex
 Then the system shall enable recording per §170.207(n)(1) standard
 And the system shall support administrative sex categories

Scenario: Record patient's sex (after December 31, 2025)
 Given the current date is after December 31, 2025
 When I record the patient's sex
 Then the system shall enable recording per §170.207(n)(2) standard
 And the system shall support updated sex categories

----- Sex Parameter for Clinical Use (Required by January 1, 2026) -----

Scenario: Record sex parameter for clinical use
 Given the current date is on or after January 1, 2026
 When I record clinical observations requiring sex parameter
 Then the system shall enable recording at least one sex parameter for clinical use
 And the recording shall be per §170.207(n)(3) standard
 And the parameter shall be context-specific for clinical use

Scenario: Support multiple sex parameters for clinical use
 Given the current date is on or after January 1, 2026
 And different clinical contexts require different parameters
 When I record sex parameters for various clinical purposes
 Then the system shall enable recording multiple sex parameters
 And each parameter shall be associated with its clinical context

----- Sexual Orientation -----

Scenario: Record patient's sexual orientation (until December 31, 2025)
 Given the current date is before December 31, 2025
 When I record the patient's sexual orientation
 Then the system shall enable recording per §170.207(o)(1) standard
 And the system shall use LOINC and SNOMED CT codes

Scenario: Record patient's sexual orientation (after December 31, 2025)
 Given the current date is after December 31, 2025
 When I record the patient's sexual orientation
 Then the system shall enable recording per §170.207(o)(3) standard
 And the system shall support updated vocabulary

Scenario: Record patient decline to specify sexual orientation
Given I am recording patient demographics
When the patient declines to specify sexual orientation
Then the system shall enable recording that patient declines to specify
And this shall be stored as distinct from missing data

----- Gender Identity -----

Scenario: Record patient's gender identity (until December 31, 2025)
Given the current date is before December 31, 2025
When I record the patient's gender identity
Then the system shall enable recording per §170.207(o)(2) standard
And the system shall use standard gender identity vocabulary

Scenario: Record patient's gender identity (after December 31, 2025)
Given the current date is after December 31, 2025
When I record the patient's gender identity
Then the system shall enable recording per §170.207(o)(3) standard
And the system shall support updated vocabulary

Scenario: Record patient decline to specify gender identity
Given I am recording patient demographics
When the patient declines to specify gender identity
Then the system shall enable recording that patient declines to specify
And this shall be stored as distinct from missing data

----- Name to Use (Required by January 1, 2026) -----

Scenario: Record patient's preferred name to use
Given the current date is on or after January 1, 2026
When I record the patient's preferred name
Then the system shall enable recording at least one preferred name to use
And the preferred name shall be distinct from legal name
And the preferred name shall be usable in clinical workflows

Scenario: Display patient's preferred name in user interface
Given the current date is on or after January 1, 2026
And the patient has a preferred name recorded
When displaying patient information
Then the system should prominently display the preferred name
And the legal name shall remain accessible when needed

----- Pronouns (Required by January 1, 2026) -----

Scenario: Record patient's pronouns
Given the current date is on or after January 1, 2026
When I record the patient's pronouns
Then the system shall enable recording at least one pronoun
And the recording shall be per §170.207(o)(4) standard
And the pronouns shall use standard vocabulary

Scenario: Support multiple pronoun sets
Given the current date is on or after January 1, 2026
And a patient uses different pronouns in different contexts

When I record the patient's pronouns
Then the system shall enable recording multiple pronoun sets
And each pronoun set shall be stored and retrievable

----- *Date of Birth* -----

Scenario: Record patient's complete date of birth
Given I am recording patient demographics
When I record the patient's date of birth
Then the system shall enable recording year, month, and day
And the date shall be stored in standard format
And the date shall be used for age calculations

Scenario: Handle unknown date of birth
Given the patient's exact date of birth is unknown
When I record demographic information
Then the system shall enable recording null value for unknown date of birth
And the system shall distinguish unknown from missing data

----- *Change and Access Demographics* -----

Scenario: Change patient demographic data
Given patient demographic data has been recorded
When demographic information changes or needs correction
Then the system shall enable changing all demographic data elements
And changes shall be audited with timestamp and user
And history of changes shall be maintained

Scenario: Access patient demographic data
Given patient demographic data has been recorded
When I need to view patient demographics
Then the system shall enable access to all demographic data elements
And demographic data shall be displayed clearly
And demographic data shall be available throughout clinical workflows

----- *Inpatient-Only: Mortality Data* -----

@inpatient-only

Scenario: Record preliminary cause of death in inpatient setting
Given I am authenticated in an inpatient setting
And a patient mortality has occurred
When I record mortality information
Then the system shall enable recording preliminary cause of death
And the cause shall be coded appropriately
And the information shall be available for death certificate

@inpatient-only

Scenario: Record date of death in inpatient setting
Given I am authenticated in an inpatient setting
And a patient mortality has occurred
When I record mortality information
Then the system shall enable recording date of death
And the date shall include date and time
And the date shall be stored in standard format

@inpatient-only

Scenario: Change mortality data in inpatient setting

Given mortality data has been recorded

When corrections are needed to mortality information

Then the system shall enable changing preliminary cause of death

And the system shall enable changing date of death

And all changes shall be audited

@inpatient-only

Scenario: Access mortality data in inpatient setting

Given mortality data has been recorded

When generating death certificate or reports

Then the system shall enable access to preliminary cause of death

And the system shall enable access to date of death

And the data shall be formatted appropriately for reporting

5.3 Screenshots

The screenshot displays a web-based data entry interface for patient demographics. The interface is titled 'Honeycomb' and includes a user profile 'demouser' and a 'LOGOUT' link. The form is organized into sections: 'Name' with fields for 'Given Name(s)' (containing 'Demo') and 'Family Name' (containing 'GraphicsChanged'); 'Identifier' with a 'Medical Record Number' field (containing 'baseehr-a5-1783149582995'); and 'Demographics' with fields for 'Birth Date' (02/29/1980), 'Birth Sex' (Female), 'Language' (Spanish (Mexico)), 'Race' (Asian), 'Ethnicity' (Karyotype), and 'Karyotype'. The form also features 'EDIT' and 'DELETE' buttons at the top right.

Figure 5.1: Patient Demographics - Data Entry Interface

Chapter 6

§ 170.315(a)(9) - Clinical Decision Support (EXPIRED)

6.1 Criterion Information

6.1.1 Maturity Level

IMPLEMENTED (EXPIRED)

6.1.2 Expiration Notice

CRITICAL: This criterion EXPIRED on January 1, 2025 per 45 CFR § 170.315(a)(9)(vi).

This criterion has been replaced by § 170.315(b)(11) - Decision Support Interventions (see Chapter 9).

Due to government website unavailability as of January 2025, it is unclear whether Base EHR certifications issued before the expiration date can still use this criterion (grandfathered), or if all certifications must now use § 170.315(b)(11).

6.1.3 Regulatory Text

From 45 CFR § 170.315(a)(9):

(9) Clinical decision support (CDS) —

(i) CDS intervention interaction. Interventions provided to a user must occur when a user is interacting with technology.

(Full regulatory text abbreviated. See BDD specification for complete requirements.)

(vi) Expiration of criterion. The adoption of this criterion for purposes of the ONC Health IT Certification Program expires on January 1, 2025.

6.2 Behavior-Driven Development Specification

Listing 6.1: 170.315(a)(9) - Clinical Decision Support

```
# /Volumes/SonicMagic/Code/honeycomb-public-release/certification/bdd/170.315-a-9-
  clinical-decision-support.feature
# §170.315(a)(9) - Clinical Decision Support (CDS)
# NOTE: This criterion expires January 1, 2025

# REGULATORY TEXT FROM 45 CFR §170.315(a)(9)
#
# (9) Clinical decision support (CDS)
#
# (i) CDS intervention interaction. Interventions provided to a user must occur when a
  user is interacting with technology.
#
# (ii) CDS configuration.
#
# (A) Enable interventions and reference resources specified in paragraphs (a)(9)(iii)
  and (iv) of this section to be configured by a limited set of identified users (e.g.,
  system administrator) based on a user's role.
#
# (B) Enable interventions:
#
# (1) Based on the following data:
```

```

#
# (i) Problem list;
#
# (ii) Medication list;
#
# (iii) Allergy and intolerance list;
#
# (iv) At least one demographic specified in paragraph (a)(5)(i) of this section;
#
# (v) Laboratory tests; and
#
# (vi) Vital signs.
#
# (2) When a patient's medications, allergies and intolerance, and problems are
#       incorporated from a transition of care/referral summary received and pursuant to
#       paragraph (b)(2)(iii)(D) of this section.
#
# (iii) Evidence-based decision support interventions. Enable a limited set of identified
#       users to select (i.e., activate) electronic CDS interventions (in addition to drug-
#       drug and drug-allergy contraindication checking) based on each one and at least one
#       combination of the data referenced in paragraphs (a)(9)(ii)(B)(1)(i) through (vi) of
#       this section.
#
# (iv) Linked referential CDS.
#
# (A) Identify for a user diagnostic and therapeutic reference information in accordance
#       at least one of the following standards and implementation specifications:
#
# (1) The standard and implementation specifications specified in §170.204(b)(3).
#
# (2) The standard and implementation specifications specified in §170.204(b)(4).
#
# (B) For paragraph (a)(9)(iv)(A) of this section, technology must be able to identify
#       for a user diagnostic or therapeutic reference information based on each one and at
#       least one combination of the data referenced in paragraphs (a)(9)(ii)(B)(1)(i), (ii),
#       and (iv) of this section.
#
# (v) Source attributes. Enable a user to review the attributes as indicated for all CDS
#       resources:
#
# (A) For evidence-based decision support interventions under paragraph (a)(9)(iii) of
#       this section:
#
# (1) Bibliographic citation of the intervention (clinical research/guideline);
#
# (2) Developer of the intervention (translation from clinical research/guideline);
#
# (3) Funding source of the intervention development technical implementation; and
#
# (4) Release and, if applicable, revision date(s) of the intervention or reference
#       source.
#
# (B) For linked referential CDS in paragraph (a)(9)(iv) of this section and drug-drug,
#       drug-allergy interaction checks in paragraph (a)(4) of this section, the developer of

```

the intervention, and where clinically indicated, the bibliographic citation of the intervention (clinical research/guideline).

#

(vi) Expiration of criterion. The adoption of this criterion for purposes of the ONC Health IT Certification Program expires on January 1, 2025.

@170.315-a-9 @cds @clinical-decision-support @expires-2025 @clinical

Feature: Clinical Decision Support (CDS)

As a healthcare provider

I want evidence-based clinical decision support

So that I can make informed treatment decisions

Background:

Given I am authenticated as a provider

And a patient record is selected

----- CDS Intervention Interaction -----

Scenario: CDS interventions occur during user interaction

Given I am actively interacting with the health IT system

When clinical decision support is triggered

Then interventions shall be provided during my interaction with technology

And interventions shall not be delayed until after interaction ends

----- CDS Configuration -----

Scenario: Configure CDS interventions based on user role

Given I am a system administrator or authorized configurator

When I configure clinical decision support interventions

Then the system shall enable configuration by limited set of identified users

And configuration shall be based on user role

And configuration capabilities shall be restricted appropriately

Scenario: Configure CDS reference resources based on user role

Given I am a system administrator or authorized configurator

When I configure reference resources for CDS

Then the system shall enable configuration by limited set of identified users

And configuration shall be based on user role

----- CDS Based on Patient Data -----

Scenario: Enable interventions based on problem list

Given the patient has documented problems

When CDS evaluates patient data

Then the system shall enable interventions based on problem list data

Scenario: Enable interventions based on medication list

Given the patient has active medications

When CDS evaluates patient data

Then the system shall enable interventions based on medication list data

Scenario: Enable interventions based on allergy and intolerance list

Given the patient has documented allergies and intolerances

When CDS evaluates patient data

Then the system shall enable interventions based on allergy and intolerance list

Scenario: Enable interventions based on demographics

Given the patient has demographic data recorded

When CDS evaluates patient data

Then the system shall enable interventions based on at least one demographic

And demographic data may include age, sex, race, ethnicity, etc.

Scenario: Enable interventions based on laboratory tests

Given the patient has laboratory test results

When CDS evaluates patient data

Then the system shall enable interventions based on laboratory test data

Scenario: Enable interventions based on vital signs

Given the patient has vital signs recorded

When CDS evaluates patient data

Then the system shall enable interventions based on vital signs data

Scenario: Enable interventions from incorporated transition of care data

Given medications, allergies, and problems are incorporated from transition of care summary

When the data is incorporated per §170.315(b)(2)(iii)(D)

Then the system shall enable interventions based on incorporated data

And interventions shall trigger automatically upon incorporation

----- Evidence-Based Decision Support Interventions -----

Scenario: Select evidence-based CDS interventions

Given I am an authorized user selecting CDS interventions

When I activate electronic CDS interventions

Then the system shall enable selection based on each individual data element

And the system shall enable selection based on combinations of data elements

And data elements include problem list, medication list, allergies, demographics, lab tests, and vital signs

Scenario: Activate CDS intervention based on problem list

Given I am selecting CDS interventions

When I activate an intervention based on problem list

Then the intervention shall utilize problem list data

And the intervention shall provide evidence-based guidance

Scenario: Activate CDS intervention based on multiple data elements

Given I am selecting CDS interventions

When I activate an intervention based on combination of data elements

Then the intervention shall utilize all specified data elements

And the intervention shall provide integrated clinical guidance

----- Linked Referential CDS -----

Scenario: Identify diagnostic reference information via Infobutton

Given I need diagnostic reference information

When I access linked referential CDS

Then the system shall identify information per §170.204(b)(3) standard

And information shall be context-specific to clinical scenario

Scenario: Identify therapeutic reference information via Infobutton

Given I need therapeutic reference information

When I access linked referential CDS

Then the system shall identify information per §170.204(b)(4) standard

And information shall be context-specific to clinical scenario

Scenario: Identify reference information based on problem list

Given the patient has documented problems

When I access linked referential CDS

Then the system shall identify information based on problem list data

Scenario: Identify reference information based on medication list

Given the patient has active medications

When I access linked referential CDS

Then the system shall identify information based on medication list data

Scenario: Identify reference information based on demographics

Given the patient has demographic data

When I access linked referential CDS

Then the system shall identify information based on demographic data

Scenario: Identify reference information based on combination of data

Given multiple types of patient data are available

When I access linked referential CDS

Then the system shall identify information based on combinations of data

And combinations may include problem list, medications, and demographics

----- Source Attributes for CDS -----

Scenario: Display source attributes for evidence-based interventions

Given an evidence-based CDS intervention is active

When I review the intervention source information

Then the system shall enable review of bibliographic citation

And the system shall display developer of the intervention

And the system shall display funding source of intervention development

And the system shall display release and revision dates

Scenario: Display source attributes for drug interaction checks

Given drug-drug or drug-allergy interaction check is displayed

When I review the intervention source information

Then the system shall display developer of the intervention

And where clinically indicated, the system shall display bibliographic citation

Scenario: Display source attributes for linked referential CDS

Given linked referential CDS is being utilized

When I review the reference source information

Then the system shall display developer of the intervention

And where clinically indicated, the system shall display bibliographic citation

Scenario: Access complete source attribute information

Given CDS interventions are configured in the system

When I need to review source attributes

Then all source attributes shall be accessible

And source attributes shall be displayed in user-friendly format
And source attributes shall support evidence-based decision making

6.3 Screenshots

Screenshot Placeholder: CDS Configuration Interface

Figure 6.1: Clinical Decision Support - Configuration Interface

Chapter 7

§ 170.315(b)(1) - Transitions of Care

7.1 Criterion Information

7.1.1 Maturity Level

VERIFIED (behavioral tests green)

7.1.2 Regulatory Text

From 45 CFR § 170.315(b)(1):

(1) Transitions of care.

(i) Send and receive via Edge Protocol. Send and receive transition-of-care/referral summaries formatted per the Consolidated CDA (C-CDA) Release 2.1, using an Edge Protocol.

(ii) Validate and display. Validate received C-CDA documents against the standard and display their contents.

(iii) Create. Create a transition-of-care/referral summary as a C-CDA document, populated from the patient's record per the Common Clinical Data Set / USCDI.

(Full regulatory text abbreviated. See BDD specification for complete requirements.)

7.1.3 Implementation

The behavioral test verifies create, validate, access, and receive: a C-CDA R2.1-enveloped transition-of-care document (Transfer Summary, LOINC 18761-7) is generated with the US Realm header, CCD template identifiers, and LOINC-coded Allergies/Medications/Problems sections, **populated from the real patient record** (the document gathers Patients, AllergyIntolerances, Medication-Requests/Statements, Conditions, Observations, and Procedures for the patient); it is persisted as a FHIR DocumentReference with the XML as a base64 attachment plus an AuditEvent; and a malformed document is rejected by the validator.

An **inbound receive** path (`clinicalDocuments.receiveCCDA`) parses an externally-authored C-CDA (via XML parsing), validates the envelope, extracts patient demographics and the section list, and stores it for display in the Clinical Documents list/detail; a Receive C-CDA upload control is provided on the export page. Structured-entry *incorporation* (creating Conditions/etc. from parsed sections) and schematron-grade validation are flagged follow-ons.

7.2 Behavior-Driven Development Specification

Listing 7.1: 170.315(b)(1) - Transitions of Care

```
# certification/bdd/170.315-b-1-transitions-of-care.feature
# §170.315(b)(1) - Transitions of Care
# BDD Format using Cucumber/Gherkin Syntax

# REGULATORY TEXT FROM 45 CFR §170.315(b)(1)
#
# (1) Transitions of care
#
# (i) Send and receive via edge protocol.
#
```

```

# (A) Send transition of care/referral summaries through a method that conforms to the
#       standard specified in §170.202(d) and that leads to such summaries being processed by
#       a service that has implemented the standard specified in §170.202(a)(2); and
#
# (B) Receive transition of care/referral summaries through a method that conforms to the
#       standard specified in §170.202(d) from a service that has implemented the standard
#       specified in §170.202(a)(2).
#
# (C) XDM processing. Receive and make available the contents of a XDM package formatted
#       in accordance with the standard adopted in §170.205(p)(1) when the technology is also
#       being certified using an SMTP-based edge protocol.
#
# (ii) Validate and display
#
# (A) Validate C-CDA conformance system performance. Demonstrate the ability to detect
#       valid and invalid transition of care/referral summaries received and formatted in
#       accordance with the standards specified in §170.205(a)(3), (4), and (5) for the
#       Continuity of Care Document, Referral Note, and (inpatient setting only) Discharge
#       Summary document templates. This includes the ability to:
#
# (1) Parse each of the document types.
#
# (2) Detect errors in corresponding "document-templates," "section-templates," and "
#       entry-templates," including invalid vocabulary standards and codes not specified in
#       the standards adopted in §170.205(a)(3), (4), and (5).
#
# (3) Identify valid document-templates and process the data elements required in the
#       corresponding section-templates and entry-templates from the standards adopted in §
#       170.205(a)(3), (4), and (5).
#
# (4) Correctly interpret empty sections and null combinations.
#
# (5) Record errors encountered and allow a user through at least one of the following
#       ways to:
#
# (i) Be notified of the errors produced.
#
# (ii) Review the errors produced.
#
# (B) Display. Display in human readable format the data included in transition of care/
#       referral summaries received and formatted according to the standards specified in §
#       170.205(a)(3), (4), and (5).
#
# (C) Display section views. Allow for the individual display of each section (and the
#       accompanying document header information) that is included in a transition of care/
#       referral summary received and formatted in accordance with the standards adopted in §
#       170.205(a)(3), (4), and (5) in a manner that enables the user to:
#
# (1) Directly display only the data within a particular section;
#
# (2) Set a preference for the display order of specific sections; and
#
# (3) Set the initial quantity of sections to be displayed.
#

```

```

# (iii) Create. Enable a user to create a transition of care/referral summary formatted
#   in accordance with the standard specified in §170.205(a)(3), (4), and (5) using the
#   Continuity of Care Document, Referral Note, and (inpatient setting only) Discharge
#   Summary document templates that includes, at a minimum:
#
# (A)
#
# (1) The data classes expressed in the standards in §170.213 and in accordance with §
#   170.205(a)(4), (5), and paragraphs (b)(1)(iii)(A)(3)(i) through (iii) of this section
#   for the time period up to and including December 31, 2025, or
#
# (2) The data classes expressed in the standards in §170.213 and in accordance with §
#   170.205(a)(4), (6), and paragraphs (b)(1)(iii)(A)(3)(i) through (iii) of this section
#   , and
#
# (3) The following data classes:
#
# (i) Assessment and plan of treatment. In accordance with the "Assessment and Plan
#   Section (V2)" of the standard specified in §170.205(a)(4); or in accordance with the
#   "Assessment Section (V2)" and "Plan of Treatment Section (V2)" of the standard
#   specified in §170.205(a)(4).
#
# (ii) Goals. In accordance with the "Goals Section" of the standard specified in §
#   170.205(a)(4).
#
# (iii) Health concerns. In accordance with the "Health Concerns Section" of the standard
#   specified in §170.205(a)(4).
#
# (iv) Unique device identifier(s) for a patient's implantable device(s). In accordance
#   with the "Product Instance" in the "Procedure Activity Procedure Section" of the
#   standard specified in §170.205(a)(4).
#
# (B) Encounter diagnoses. Formatted according to at least one of the following standards
#   :
#
# (1) The standard specified in §170.207(i).
#
# (2) At a minimum, the version of the standard specified in §170.207(a)(1).
#
# (C) Cognitive status.
#
# (D) Functional status.
#
# (E) Ambulatory setting only. The reason for referral; and referring or transitioning
#   provider's name and office contact information.
#
# (F) Inpatient setting only. Discharge instructions.
#
# (G) Patient matching data. First name, last name, previous name, middle name (including
#   middle initial), suffix, date of birth, current address, phone number, and sex. The
#   following constraints apply:
#
# (1) Date of birth constraint.
#

```

```

# (i) The year, month and day of birth must be present for a date of birth. The
    technology must include a null value when the date of birth is unknown.
#
# (ii) Optional. When the hour, minute, and second are associated with a date of birth
    the technology must demonstrate that the correct time zone offset is included.
#
# (2) Phone number constraint. Represent phone number (home, business, cell) in
    accordance with the standards adopted in §170.207(q)(1). All phone numbers must be
    included when multiple phone numbers are present.
#
# (3) Sex Constraint: Represent sex with the standard adopted in §170.207(n)(1) up to and
    including December 31, 2025; or with the standard adopted in §170.207(n)(2).

@170.315-b-1 @transitions-of-care @toc @care-coordination
Feature: Transitions of Care
  As a healthcare provider
  I want to send and receive comprehensive transition of care summaries
  So that I can ensure care continuity when patients move between care settings

Background:
  Given I am authenticated as a provider
  And a patient record is selected

# ----- Send Transition of Care Summary via Edge Protocol -----

Scenario: Send transition of care summary via Direct
  Given a patient is being transitioned to another provider
  When I create and send a transition of care summary
  Then the system shall send through method conforming to §170.202(d)
  And the summary shall be processed by service implementing §170.202(a)(2)
  And the transmission shall use secure messaging

Scenario: Receive transition of care summary via Direct
  Given another provider is sending a transition of care summary for my patient
  When the summary is transmitted
  Then the system shall receive through method conforming to §170.202(d)
  And the system shall receive from service implementing §170.202(a)(2)
  And the received summary shall be available for processing

Scenario: Process XDM package when using SMTP-based edge protocol
  Given the technology uses SMTP-based edge protocol for Direct messaging
  When receiving an XDM package
  Then the system shall receive XDM package formatted per §170.205(p)(1)
  And the system shall make available the contents of the XDM package
  And contents shall be accessible for clinical review

# ----- Validate C-CDA Conformance -----

Scenario: Detect valid transition of care summary
  Given a transition of care summary is received
  When the system validates the document
  Then the system shall detect valid summaries formatted per §170.205(a)(3), (4), (5)
  And valid documents shall be processed for display and reconciliation

```


Scenario: Detect invalid transition of care summary

Given a malformed transition of care summary is received
When the system validates the document
Then the system shall detect invalid summaries
And errors shall be identified and reported

Scenario: Parse CCD document template

Given a Continuity of Care Document is received
When the system processes the document
Then the system shall parse the CCD template
And all sections shall be accessible

Scenario: Parse Referral Note document template

Given a Referral Note is received
When the system processes the document
Then the system shall parse the Referral Note template
And all sections shall be accessible

@inpatient-only

Scenario: Parse Discharge Summary document template

Given I am in an inpatient setting
And a Discharge Summary is received
When the system processes the document
Then the system shall parse the Discharge Summary template
And all sections shall be accessible

Scenario: Detect errors in document-templates

Given a transition of care summary with template errors is received
When the system validates document structure
Then the system shall detect errors in document-templates
And specific template errors shall be identified

Scenario: Detect errors in section-templates

Given a transition of care summary with section errors is received
When the system validates document structure
Then the system shall detect errors in section-templates
And specific section errors shall be identified

Scenario: Detect errors in entry-templates

Given a transition of care summary with entry errors is received
When the system validates document structure
Then the system shall detect errors in entry-templates
And specific entry errors shall be identified

Scenario: Detect invalid vocabulary standards

Given a transition of care summary with invalid codes is received
When the system validates vocabulary
Then the system shall detect invalid vocabulary standards
And invalid codes shall be identified

Scenario: Detect codes not specified in standards

Given a transition of care summary with non-standard codes is received
When the system validates vocabulary
Then the system shall detect codes not specified in §170.205(a)(3), (4), (5)

And non-standard codes shall be reported

Scenario: Identify valid document-templates

Given a properly formatted transition of care summary is received

When the system validates the document

Then the system shall identify valid document-templates

And valid templates shall be processed successfully

Scenario: Process required data elements

Given a valid transition of care summary is received

When the system extracts data elements

Then the system shall process data elements required in corresponding section-templates

And the system shall process data elements required in corresponding entry-templates

And all required data per §170.205(a)(3), (4), (5) shall be extracted

Scenario: Correctly interpret empty sections

Given a transition of care summary with empty sections is received

When the system processes the document

Then the system shall correctly interpret empty sections

And empty sections shall not cause processing errors

Scenario: Correctly interpret null combinations

Given a transition of care summary with null values is received

When the system processes the document

Then the system shall correctly interpret null combinations

And null values shall be handled appropriately

Scenario: Record errors encountered during validation

Given a transition of care summary with multiple errors is received

When the system validates the document

Then the system shall record errors encountered

And errors shall be stored for review

Scenario: Notify user of validation errors

Given validation errors have been recorded

When errors are detected

Then the system shall enable user notification of errors produced

And notifications shall be timely and clear

Scenario: Enable user to review validation errors

Given validation errors have been recorded

When a user needs to review errors

Then the system shall enable user to review errors produced

And error details shall be accessible

And errors shall be presented in actionable format

----- Display Received Transition of Care Summary -----

Scenario: Display transition of care summary in human readable format

Given a valid transition of care summary is received

When I view the summary

Then the system shall display data in human readable format

And the display shall be formatted per §170.205(a)(3), (4), (5)

And all included data shall be viewable

Scenario: Display CCD in human readable format

Given a Continuity of Care Document is received

When I view the document

Then all data shall be displayed in human readable format

And the display shall be clinically useful

Scenario: Display Referral Note in human readable format

Given a Referral Note is received

When I view the document

Then all data shall be displayed in human readable format

And referral information shall be prominent

@inpatient-only

Scenario: Display Discharge Summary in human readable format

Given I am in an inpatient setting

And a Discharge Summary is received

When I view the document

Then all data shall be displayed in human readable format

And discharge information shall be prominent

----- Display Individual Sections -----

Scenario: Display individual section from transition summary

Given a transition of care summary is received

When I want to view a specific section

Then the system shall allow individual display of each section

And accompanying document header information shall be displayed

Scenario: Directly display only data within particular section

Given a transition of care summary with multiple sections is received

When I select a specific section to view

Then the system shall enable direct display of only that section's data

And other sections shall be hidden from view

Scenario: Set preference for display order of sections

Given I have specific workflow preferences

When I configure section display settings

Then the system shall enable setting preference for display order of sections

And my preferences shall be saved and applied

Scenario: Set initial quantity of sections displayed

Given I want to control information density

When I configure section display settings

Then the system shall enable setting initial quantity of sections displayed

And I shall be able to expand to view additional sections

----- Create Transition of Care Summary -----

Scenario: Create CCD document template

Given I need to send a transition of care summary

When I create a Continuity of Care Document

Then the system shall format per §170.205(a)(3), (4), (5)

And the CCD shall include all required USCDI data classes per §170.213

Scenario: Create Referral Note document template

Given I need to send a referral

When I create a Referral Note

Then the system shall format per §170.205(a)(3), (4), (5)

And the Referral Note shall include all required USCDI data classes per §170.213

@inpatient-only

Scenario: Create Discharge Summary document template

Given I am in an inpatient setting

And a patient is being discharged

When I create a Discharge Summary

Then the system shall format per §170.205(a)(3), (4), (5)

And the Discharge Summary shall include all required USCDI data classes per §170.213

Scenario: Include all USCDI data classes in transition summary

Given I am creating a transition of care summary

When I generate the summary

Then the system shall include all data classes per §170.213

And data shall be formatted per §170.205(a)(4), (5) or (6) based on date

And all mandatory data elements shall be included

Scenario: Include assessment and plan of treatment

Given I am creating a transition of care summary

When I generate the summary

Then the system shall include assessment and plan per "Assessment_□and_□Plan_□Section_□(V2)"

Or the system shall include separate "Assessment_□Section_□(V2)" and "Plan_□of_□Treatment_□Section_□(V2)"

And content shall conform to §170.205(a)(4)

Scenario: Include goals in transition summary

Given I am creating a transition of care summary

When I generate the summary

Then the system shall include goals per "Goals_□Section"

And goals shall conform to §170.205(a)(4)

Scenario: Include health concerns in transition summary

Given I am creating a transition of care summary

When I generate the summary

Then the system shall include health concerns per "Health_□Concerns_□Section"

And health concerns shall conform to §170.205(a)(4)

Scenario: Include unique device identifiers in transition summary

Given the patient has implantable devices

When I create a transition of care summary

Then the system shall include UDIs per "Product_□Instance" in "Procedure_□Activity_□Procedure_□Section"

And UDIs shall conform to §170.205(a)(4)

Scenario: Include encounter diagnoses in transition summary

Given I am creating a transition of care summary

When I generate the summary

Then the system shall include encounter diagnoses
And diagnoses shall be formatted per §170.207(i) or §170.207(a)(1)

Scenario: Include cognitive status in transition summary

Given I am creating a transition of care summary
When I generate the summary
Then the system shall include cognitive status
And cognitive status shall be appropriately coded

Scenario: Include functional status in transition summary

Given I am creating a transition of care summary
When I generate the summary
Then the system shall include functional status
And functional status shall be appropriately coded

----- Patient Matching Data -----

Scenario: Include patient matching data in transition summary

Given I am creating a transition of care summary
When I generate the summary
Then the system shall include first name, last name, previous name, middle name,
suffix
And the system shall include date of birth
And the system shall include current address
And the system shall include phone number
And the system shall include sex

Scenario: Include complete date of birth with null for unknown

Given I am creating a transition of care summary
When including date of birth
Then year, month, and day must be present for date of birth
And the system shall include null value when date of birth is unknown

Scenario: Optionally include time zone with date of birth

Given I am creating a transition of care summary
When date of birth includes time components
Then the system may include hour, minute, and second
And if time is included, correct time zone offset must be included

Scenario: Represent phone numbers per standard

Given I am creating a transition of care summary
When including phone numbers
Then phone numbers shall be represented per §170.207(q)(1)
And all phone numbers (home, business, cell) must be included when multiple are
present

Scenario: Represent sex per standard until December 31, 2025

Given the current date is before December 31, 2025
When including sex in transition summary
Then sex shall be represented per §170.207(n)(1)

Scenario: Represent sex per updated standard after December 31, 2025

Given the current date is after December 31, 2025
When including sex in transition summary

```
Then sex shall be represented per §170.207(n)(2)

# ----- Ambulatory Setting Specific -----

@ambulatory-only
Scenario: Include reason for referral in ambulatory setting
  Given I am in an ambulatory setting
  When I create a transition of care summary
  Then the system shall include reason for referral
  And the reason shall be clinically meaningful

@ambulatory-only
Scenario: Include referring provider information in ambulatory setting
  Given I am in an ambulatory setting
  When I create a transition of care summary
  Then the system shall include referring provider's name
  And the system shall include referring provider's office contact information

# ----- Inpatient Setting Specific -----

@inpatient-only
Scenario: Include discharge instructions in inpatient setting
  Given I am in an inpatient setting
  When I create a transition of care summary
  Then the system shall include discharge instructions
  And discharge instructions shall be comprehensive
```

7.3 Screenshots

The screenshot shows the Honeycomb BaseEHR CodaTransition interface. The top navigation bar includes the Honeycomb logo, a settings icon, a document icon, and links for CLEAR PATIENT, demouser, and LOGOUT. Below the navigation bar, the patient information is displayed: ID (baseehr-b1-1783149619115), Birth Date (Mar 03, 1962), Gender (female), and Phone (Unknown).

C-CDA Document Export - ONC §170.315(b)(1)

The main content area is divided into three columns:

- Patient Selection:** Lists three patients: John Doe (MRN: 12345 | DOB: 1970-01-15), Jane Smith (MRN: 67890 | DOB: 1985-06-20), and Robert Johnson (MRN: 11223 | DOB: 1955-11-30).
- Document Configuration:** Shows the Document Type as "Summary of Episode Note (CCD)" with 14 sections. The Export Format is set to "CCD".
- Document Sections:** Lists 10 sections selected, all marked as "Required": Allergies, Medications, Problems, Procedures, and Results.
- ONC Compliance:** Shows four compliance checks, all marked as successful: §170.315(b)(1) - Transitions of Care, C-CDA R2.1 Compliant, Vocabulary Standards (LOINC, SNOMED, RxNorm), and Schematron Validation.

Figure 7.1: Transitions of Care - C-CDA Generation

Chapter 8

§ 170.315(b)(4) - Real-Time Prescription Benefit

8.1 Criterion Information

8.1.1 Maturity Level

VERIFIED (behavioral tests green)

8.1.2 Regulatory Text

From *45 CFR § 170.315(b)(4)*:

(4) Real-time prescription benefit (RTPB).

Enable a user, for a prescribed drug, to request and receive real-time prescription benefit information — including patient cost, formulary/coverage status, and any therapeutic alternatives — from a benefit source.

(Full regulatory text abbreviated. See BDD specification for complete requirements.)

8.1.3 Implementation Status

The behavioral test drives the full RTPB round-trip: it composes an `RTPBRequest` for a prescribed product, transacts it against a benefit responder (a formulary/PBM responder from the responder registry), and receives an `RTPBResponse` carrying the patient's cost and cheaper therapeutic alternatives (a brand statin returns generic substitutions). Both request and response XML are rendered and persisted with request/response linkage, and the completed check appears in the transaction history. A configurable live endpoint is supported alongside the built-in certification-evidence responders.

Future compliance date: (b)(4) is an HTI-4 criterion required by 1/1/2028. The current baseline (request/response, patient cost, alternatives, live-endpoint capability) is verified; the final standard-version binding is treated as informational until closer to the compliance date, not a gap.

8.2 Behavior-Driven Development Specification

A dedicated Gherkin feature file for (b)(4) is not yet authored; the executable specification is the behavioral Nightwatch test `certification/tdd/base_ehr/170.315.b.4.test.js`, which drives the request/response round-trip, patient-cost and therapeutic-alternative assertions, and the transaction-history retrieval described above.

8.3 Screenshots

The screenshot displays the Honeycomb BaseEHR RtpbCheck interface. At the top, a dark blue header contains the Honeycomb logo, a hamburger menu, and user controls (CLEAR PATIENT, demouser, LOGOUT). Below the header, a patient information bar shows the ID (baseehr-b4-1783135893772), Birth Date (Aug 08, 1948), Gender (male), and Phone (Unknown). The main content area is titled "Prescription Benefit" and "Real-Time Prescription Benefit - § 170.315(b)(4)". It includes a "Responder" dropdown menu set to "Sample PBM Plan" and a "Formulary" button. Below this, a "Transaction History" section shows 1 transaction(s). The transaction is detailed in a table with columns: Date, Product, Out-of-pocket, Alternatives, Mode, and Status.

Date	Product	Out-of-pocket	Alternatives	Mode	Status
2026-07-04 03:31:43	Lipitor 20 MG Oral Tablet	\$96.10	3	mock	completed

Figure 8.1: Real-Time Prescription Benefit - Cost and Alternatives

Chapter 9

§ 170.315(b)(11) - Decision Support Interventions

9.1 Criterion Information

9.1.1 Maturity Level

VERIFIED (behavioral tests green)

9.1.2 Replacement Notice

This criterion REPLACES § 170.315(a)(9) which expired January 1, 2025.

Decision Support Interventions (DSI) is an enhanced version of the legacy Clinical Decision Support criterion, adding requirements for:

- Predictive decision support interventions (AI/ML algorithms)
- Demographic data usage transparency
- Electronic intervention feedback mechanism
- Enhanced risk and fairness disclosures for predictive DSI

9.1.3 Regulatory Text

From *45 CFR § 170.315(b)(11)*:

(11) Decision Support Interventions.

(Full regulatory text includes all requirements from a-9 plus enhanced requirements for predictive interventions, demographic transparency, feedback mechanisms, and algorithm governance. See BDD specification for complete requirements.)

9.2 Behavior-Driven Development Specification

Listing 9.1: 170.315(b)(11) - Decision Support Interventions

```
# /Volumes/SonicMagic/Code/honeycomb-public-release/certification/bdd/170.315-b-11-
  decision-support-interventions.feature
# §170.315(b)(11) - Decision Support Interventions

# REGULATORY TEXT FROM 45 CFR §170.315(b)(11)
#
# (11) Use of date of birth as expressed in the standards in §170.213;

@170.315-b-11 @dsi @predictive-ai @algorithm-transparency @care-coordination
Feature: Decision Support Interventions
  As a healthcare provider
  I want transparent and fair decision support interventions
  So that I can make evidence-based decisions while understanding AI/algorithm influences

Background:
  Given I am authenticated as a provider
  And I have a patient record open
  And decision support interventions are configured
```

----- *Intervention Interaction* -----

Scenario: Interventions occur during user interaction

Given I am actively interacting with health IT

When decision support is triggered

Then interventions shall be provided during my interaction with technology

And interventions shall occur at clinically appropriate times

----- *Configuration* -----

Scenario: Configure DSI interventions based on user role

Given I am authorized to configure decision support

When configuring interventions

Then the system shall enable configuration by limited set of identified users

And configuration shall be based on user role

And configuration capabilities shall be appropriately restricted

Scenario: Enable interventions when data incorporated from transition of care

Given medications, allergies, and problems are incorporated from transition summary

When data is incorporated per §170.315(b)(2)(iii)(D)

Then interventions shall be enabled based on incorporated data

And interventions shall trigger appropriately

Scenario: Provide electronic feedback capability for interventions

Given I am using evidence-based decision support interventions

When I want to provide feedback on intervention

Then the system shall enable me to provide electronic feedback data

And feedback shall be collectible for selected interventions

And limited set of users shall be able to export feedback in computable format

Scenario: Export intervention feedback data

Given electronic feedback has been collected

When authorized users need to analyze feedback

Then feedback data shall be exportable in computable format

And export shall include intervention, action taken, user feedback, user, date, and location

And feedback analysis shall support intervention improvement

----- *Evidence-Based Decision Support Intervention Selection* -----

Scenario: Select evidence-based interventions using USCDI data

Given I am authorized to select decision support interventions

When activating evidence-based interventions

Then interventions may use problems per §170.213

And interventions may use medications per §170.213

And interventions may use allergies and intolerances per §170.213

And interventions may use demographics per §170.213

And interventions may use laboratory data per §170.213

And interventions may use vital signs per §170.213

And interventions may use unique device identifiers per §170.213

And interventions may use procedures per §170.213

----- *Predictive Decision Support Intervention Selection* -----

```
Scenario: Select predictive interventions using any USCDI data
  Given I am authorized to select decision support interventions
  When activating predictive decision support interventions
  Then interventions may use any data expressed in §170.213
  And predictive algorithms shall utilize appropriate data elements
  And data inputs shall be transparent

# ----- Source Attributes for Evidence-Based Interventions -----

Scenario: Display evidence-based intervention source attributes
  Given an evidence-based intervention is active
  When I review intervention source information
  Then the system shall support display of bibliographic citation
  And the system shall support display of intervention developer
  And the system shall support display of funding source for development
  And the system shall support display of release and revision dates

Scenario: Display demographic data usage in evidence-based intervention
  Given an evidence-based intervention uses demographic data
  When I review intervention details
  Then the system shall indicate use of race per §170.213
  And the system shall indicate use of ethnicity per §170.213
  And the system shall indicate use of language per §170.213
  And the system shall indicate use of sexual orientation per §170.213
  And the system shall indicate use of gender identity per §170.213
  And the system shall indicate use of sex per §170.213
  And the system shall indicate use of date of birth per §170.213

Scenario: Display SDOH and health status data usage
  Given an evidence-based intervention uses SDOH data
  When I review intervention details
  Then the system shall indicate use of SDOH data per §170.213
  And the system shall indicate use of health status assessments data per §170.213

# ----- Source Attributes for Predictive Interventions (EXTENSIVE) -----

Scenario: Display predictive intervention details and output
  Given a predictive decision support intervention is active
  When I review intervention details
  Then the system shall display name and contact information for developer
  And the system shall display funding source of technical implementation
  And the system shall display description of value produced as output
  And the system shall display whether output is prediction, classification,
    recommendation, evaluation, analysis, or other type

Scenario: Display predictive intervention purpose
  Given a predictive intervention is active
  When I review intervention purpose
  Then the system shall display intended use of intervention
  And the system shall display intended patient population(s)
  And the system shall display intended user(s)
  And the system shall display intended decision-making role (informs, augments,
    replaces clinical management)
```

Scenario: Display cautioned out-of-scope use

Given a predictive intervention has usage limitations

When I review intervention limitations

Then the system shall display description of cautioned tasks, situations, or populations

And the system shall display known risks, inappropriate settings, inappropriate uses, or limitations

Scenario: Display intervention development details and input features

Given a predictive intervention is active

When I review development details

Then the system shall display exclusion and inclusion criteria for training data

And the system shall display use of demographic variables as input features

And the system shall display description of demographic representativeness in training data

And the system shall display description of relevance of training data to deployed setting

Scenario: Display fairness process for intervention development

Given a predictive intervention is active

When I review fairness information

Then the system shall display description of approach to ensure intervention output is fair

And the system shall display description of approaches to manage, reduce, or eliminate bias

Scenario: Display external validation process

Given a predictive intervention has been externally validated

When I review validation information

Then the system shall display description of data source, setting, or environment for validation

And the system shall display party that conducted external testing

And the system shall display demographic representativeness of external data

And the system shall display description of external validation process

Scenario: Display quantitative performance measures

Given a predictive intervention is active

When I review performance metrics

Then the system shall display validity in test data from training source

And the system shall display fairness in test data from training source

And the system shall display validity in external data

And the system shall display fairness in external data

And the system shall display references to evaluation of outcomes impact

Scenario: Display ongoing maintenance information

Given a predictive intervention is deployed

When I review maintenance information

Then the system shall display process and frequency for monitoring validity over time

And the system shall display validity in local data

And the system shall display process and frequency for monitoring fairness over time

And the system shall display fairness in local data

Scenario: Display update and validation schedule

Given a predictive intervention is in use

When I review update information

Then the system shall display process and frequency for intervention updates

And the system shall display frequency for performance correction when risks identified

Scenario: Indicate when optional information not available

Given a predictive intervention source attributes

When certain information is not available for review

Then the system shall indicate when information not available for external validation

And the system shall indicate when information not available for external validity metrics

And the system shall indicate when information not available for external fairness metrics

And the system shall indicate when information not available for outcome evaluations

And the system shall indicate when information not available for local validity

And the system shall indicate when information not available for local fairness

And the system shall indicate when information not available for update schedule details

----- Access and Modify Source Attributes -----

Scenario: Access complete plain language descriptions for developer-supplied interventions

Given interventions are supplied by health IT developer

When authorized users need to review source attributes

Then limited set of identified users shall be able to access complete descriptions

And descriptions shall be in plain language

And descriptions shall be up-to-date

Scenario: Modify source attributes for interventions

Given intervention source attributes need updating

When authorized users modify attributes

Then limited set of identified users shall record, change, and access source attributes

And modifications shall be tracked with audit trail

And all users shall see updated information

Scenario: Record additional attributes for predictive interventions

Given predictive interventions may have additional relevant attributes

When documenting intervention characteristics

Then limited set of users shall record, change, and access additional attributes

And additional attributes supplement required attributes

And comprehensive documentation shall be maintained

----- Intervention Risk Management for Predictive DSI -----

Scenario: Conduct risk analysis for predictive intervention

Given a predictive decision support intervention is implemented

When assessing intervention risks

Then intervention shall be subject to analysis of potential risks and adverse impacts

And analysis shall assess validity, reliability, robustness, fairness

And analysis shall assess intelligibility, safety, security, and privacy

Scenario: Implement risk mitigation for predictive intervention

Given risks have been identified for predictive intervention

When implementing risk controls

Then intervention shall be subject to practices to mitigate identified risks

And mitigation shall address all significant risk categories

And mitigation effectiveness shall be monitored

Scenario: Apply governance to predictive intervention

Given a predictive intervention is in production use

When managing intervention lifecycle

Then intervention shall be subject to policies and implemented controls for governance

And governance shall include how data are acquired, managed, and used

And governance shall ensure ongoing oversight and accountability

9.3 Screenshots

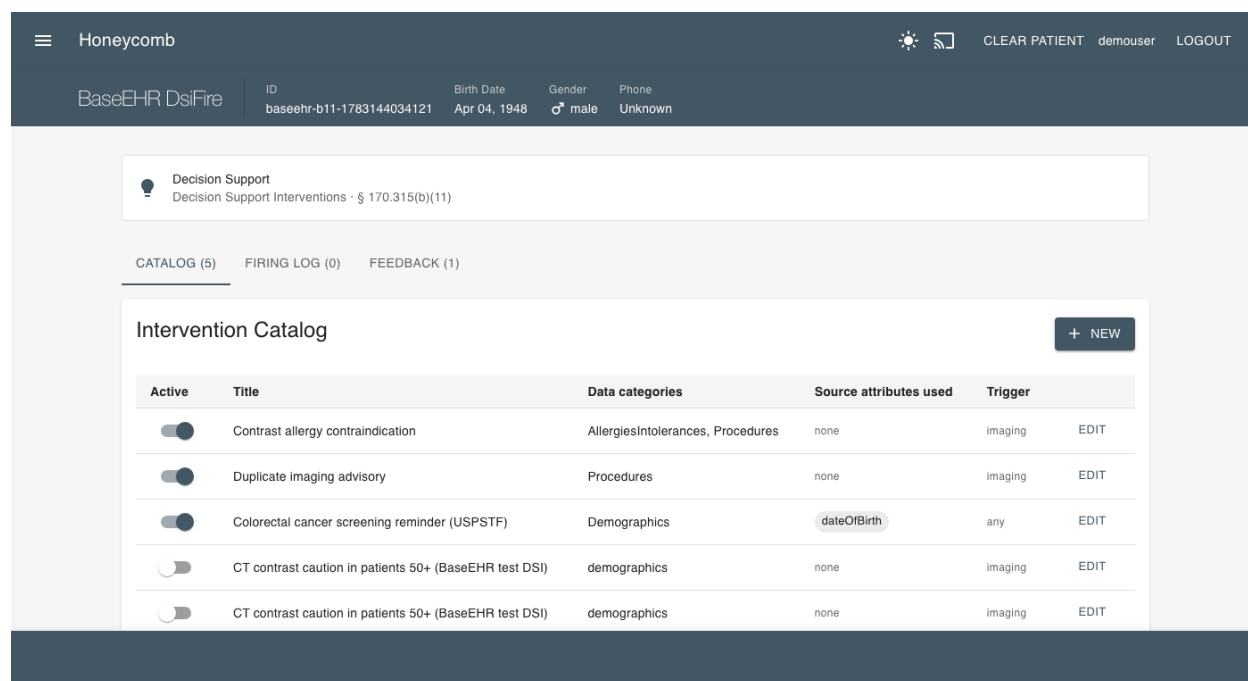


Figure 9.1: Decision Support Interventions - Configuration Interface

Chapter 10

§ 170.315(a)(14) - Implantable Device List

10.1 Criterion Information

10.1.1 Maturity Level

VERIFIED (behavioral tests green)

10.1.2 Regulatory Text

From 45 CFR § 170.315(a)(14):

(14) Implantable device list.

(i) Record Unique Device Identifiers associated with a patient's Implantable Devices.

(ii) Parse the following identifiers from a Unique Device Identifier.

(Full regulatory text abbreviated. See BDD specification for complete requirements.)

10.2 Behavior-Driven Development Specification

Listing 10.1: 170.315(a)(14) - Implantable Device List

```
# /Volumes/SonicMagic/Code/honeycomb-public-release/certification/bdd/170.315-a-14-
  implantable-device-list.feature
# §170.315(a)(14) - Implantable Device List

# REGULATORY TEXT FROM 45 CFR §170.315(a)(14)
#
# (14) Implantable device list.
#
# (i) Record Unique Device Identifiers associated with a patient's Implantable Devices.
#
# (ii) Parse the following identifiers from a Unique Device Identifier:
#
# (A) Device Identifier; and
#
# (B) The following identifiers that compose the Production Identifier:
#
# (1) The lot or batch within which a device was manufactured;
#
# (2) The serial number of a specific device;
#
# (3) The expiration date of a specific device;
#
# (4) The date a specific device was manufactured; and
#
# (5) For an HCT/P regulated as a device, the distinct identification code required by 21
    CFR 1271.290(c).
#
# (iii) Obtain and associate with each Unique Device Identifier:
#
# (A) A description of the implantable device referenced by at least one of the following
    :
#
```

```

# (1) The "GMDN PT Name" attribute associated with the Device Identifier in the Global
    Unique Device Identification Database.
#
# (2) The "SNOMED CTö Description" mapped to the attribute referenced in in paragraph (a)
    (14)(iii)(A)(1) of this section.
#
# (B) The following Global Unique Device Identification Database attributes:
#
# (1) "Brand Name";
#
# (2) "Version or Model";
#
# (3) "Company Name";
#
# (4) "What MRI safety information does the labeling contain?"; and
#
# (5) "Device required to be labeled as containing natural rubber latex or dry natural
    rubber (21 CFR 801.437).".
#
# (iv) Display to a user an implantable device list consisting of:
#
# (A) The active Unique Device Identifiers recorded for the patient;
#
# (B) For each active Unique Device Identifier recorded for a patient, the description of
    the implantable device specified by paragraph (a)(14)(iii)(A) of this section; and
#
# (C) A method to access all Unique Device Identifiers recorded for a patient.
#
# (v) For each Unique Device Identifier recorded for a patient, enable a user to access:
#
# (A) The Unique Device Identifier;
#
# (B) The description of the implantable device specified by paragraph (a)(14)(iii)(A) of
    this section;
#
# (C) The identifiers associated with the Unique Device Identifier, as specified by
    paragraph (a)(14)(ii) of this section; and
#
# (D) The attributes associated with the Unique Device Identifier, as specified by
    paragraph (a)(14)(iii)(B) of this section.
#
# (vi) Enable a user to change the status of a Unique Device Identifier recorded for a
    patient.

@170.315-a-14 @udi @implantable-devices @clinical
Feature: Implantable Device List
  As a healthcare provider
  I want to record and track implantable devices using Unique Device Identifiers
  So that I can ensure patient safety, support recall management, and provide accurate
    device information

Background:
  Given I am authenticated as a provider
  And a patient record is selected

```

----- Record UDI -----

Scenario: Record Unique Device Identifier for implantable device

Given a patient has received an implantable device

When I record the device information

Then the system shall enable recording of Unique Device Identifiers

And the UDI shall be associated with the patient record

And the UDI shall be stored in structured format

----- Parse UDI Components -----

Scenario: Parse Device Identifier from UDI

Given a Unique Device Identifier has been recorded

When the system processes the UDI

Then the system shall parse the Device Identifier (DI)

And the DI shall be stored as separate data element

Scenario: Parse Production Identifier components from UDI

Given a Unique Device Identifier has been recorded

When the system processes the UDI

Then the system shall parse lot or batch number

And the system shall parse serial number

And the system shall parse expiration date

And the system shall parse manufacture date

And all production identifiers shall be stored as separate data elements

Scenario: Parse HCT/P distinct identification code from UDI

Given a Unique Device Identifier for HCT/P regulated device has been recorded

When the system processes the UDI

Then the system shall parse distinct identification code per 21 CFR 1271.290(c)

And the code shall be stored appropriately

----- Obtain Device Information from GUDID -----

Scenario: Obtain device description from GUDID

Given a Unique Device Identifier has been entered

When the system queries the Global Unique Device Identification Database

Then the system shall obtain device description

And description may be "GMDN_PT_Name" from GUDID

Or description may be "SNOMED_CT_Description" mapped to GMDN PT Name

Scenario: Obtain device brand name from GUDID

Given a Unique Device Identifier has been entered

When the system queries GUDID

Then the system shall obtain "Brand_Name" attribute

And the brand name shall be associated with the UDI

Scenario: Obtain device version or model from GUDID

Given a Unique Device Identifier has been entered

When the system queries GUDID

Then the system shall obtain "Version_or_Model" attribute

And the version/model shall be associated with the UDI

Scenario: Obtain device company name from GUDID
Given a Unique Device Identifier has been entered
When the system queries GUDID
Then the system shall obtain "Company_Name" attribute
And the manufacturer information shall be associated with the UDI

Scenario: Obtain MRI safety information from GUDID
Given a Unique Device Identifier has been entered
When the system queries GUDID
Then the system shall obtain MRI safety information
And the information shall indicate MRI labeling content
And MRI safety information shall be prominently available

Scenario: Obtain latex allergy information from GUDID
Given a Unique Device Identifier has been entered
When the system queries GUDID
Then the system shall obtain latex labeling information per 21 CFR 801.437
And latex information shall be prominently available for patient safety

----- Display Implantable Device List -----

Scenario: Display patient's implantable device list
Given a patient has recorded Unique Device Identifiers
When I view the implantable device list
Then the system shall display active Unique Device Identifiers
And the system shall display device descriptions
And the display shall be organized and easily reviewable

Scenario: Display device description for each active UDI
Given a patient has multiple active implantable devices
When I view the implantable device list
Then for each active UDI the system shall display device description
And description shall be from GUDID per requirements
And description shall be clinically meaningful

Scenario: Provide method to access all UDIs for patient
Given a patient has both active and inactive device records
When I need to view complete device history
Then the system shall provide method to access all UDIs
And I shall be able to distinguish active from inactive devices
And complete device history shall be accessible

----- Access Individual UDI Details -----

Scenario: Access complete UDI information
Given a patient has a recorded Unique Device Identifier
When I select a specific UDI to review
Then the system shall enable access to the complete UDI
And the system shall display the device description
And the system shall display parsed production identifiers
And the system shall display GUDID attributes

Scenario: Access parsed identifiers for UDI
Given a patient has a recorded Unique Device Identifier

When I access the UDI details
 Then the system shall display lot/batch number
 And the system shall display serial number
 And the system shall display expiration date
 And the system shall display manufacture date
 And the system shall display HCT/P code if applicable

Scenario: Access GUDID attributes for UDI

Given a patient has a recorded Unique Device Identifier
 When I access the UDI details
 Then the system shall display brand name
 And the system shall display version or model
 And the system shall display company name
 And the system shall display MRI safety information
 And the system shall display latex information

----- Change UDI Status -----

Scenario: Change status of implantable device

Given a patient has a recorded Unique Device Identifier
 When the device is explanted or status changes
 Then the system shall enable changing the status of the UDI
 And status change shall be recorded with date and reason
 And device shall move from active to inactive list

10.3 Screenshots

The screenshot displays the Honeycomb BaseEHR interface for a patient's device list. The patient's information is shown at the top: ID baseehr-a14-1783131518899, Birth Date Nov 11, 1955, Gender male, and Phone Unknown. The main section is titled "Implantable Device Registry - ONC §170.315(g)(7)". It features a search bar and filters for Category (All Categories) and Class (All Classes). Below these are tabs for CATALOG, AUGMENTATIONS, and a SCAN UDI button. The main table lists five devices:

Device	Cyber	Type	Manufacturer	Class	Status	Battery	Implant Date	Actions
NeuroPulse™ Pacemaker v3.0 NP-3000	CYBER	Dual-chamber pacemaker	Arasaka Medical	Class III	active	85%	2023-06-15	VIEW
CardioGuard™ Defibrillator CG-500X	CYBER	Implantable Cardioverter Defibrillator	Millitech Biotech	Class III	active	92%	2023-08-22	VIEW
VentriFlow™ Left Ventricular Assist Device VF-HM3		Left ventricular assist device (LVAD)	Trauma Team International	Class III	active	External controller (rechargeable)	2024-03-12	VIEW
RhythmWatch™ Insertable Loop Recorder RW-ILR		Implantable cardiac monitor	Dynalar Technologies	Class II	active	88%	2023-10-02	VIEW
CortexLink™ Neural Bridge CLB-2.1	CYBER	Brain-Computer Interface	Raven Microcyber	Class III	active	Wireless charging	2024-01-10	VIEW

Figure 10.1: Implantable Device List - Patient View

Chapter 11

§ 170.315(e)(1) - View, Download, Transmit

11.1 Criterion Information

11.1.1 Maturity Level

IMPLEMENTED

11.1.2 Regulatory Text

From 45 CFR § 170.315(e)(1):

(1) View, download, and transmit to 3rd party.

The technology must enable patients to view, download, and transmit their health information.

(Full regulatory text abbreviated. See BDD specification for complete requirements.)

11.2 Behavior-Driven Development Specification

Listing 11.1: 170.315(e)(1) - View, Download, Transmit

```
# /Volumes/SonicMagic/Code/honeycomb-public-release/certification/bdd/170.315-e-1-view-
  download-transmit.feature
# §170.315(e)(1) - View, Download, and Transmit to 3rd Party

# REGULATORY TEXT FROM 45 CFR §170.315(e)(1)
#
# (1) The action(s) (i.e., view, download, transmission) that occurred;

@170.315-e-1 @vdt @patient-access @patient-engagement
Feature: View, Download, and Transmit to 3rd Party
  As a patient
  I want to view, download, and transmit my health information
  So that I can access my data and share it with others

  Background:
    Given I am authenticated as a patient
    And my health information is available in the system

  Scenario: View health information online
    Given I am accessing my patient portal
    When I view my health information
    Then the system shall display my data in human readable format
    And all available USCDI data shall be viewable
    And the information shall be current and accurate

  Scenario: Download health information
    Given I am viewing my health information
    When I choose to download my data
    Then the system shall enable download
    And download shall be in machine-readable format
    And download shall include all requested data

  Scenario: Transmit to third party via encrypted email
```

Given I want to share my health information
When I transmit to a third party email address
Then the system shall transmit encrypted
And transmission shall be secure
And third party shall be able to receive the data

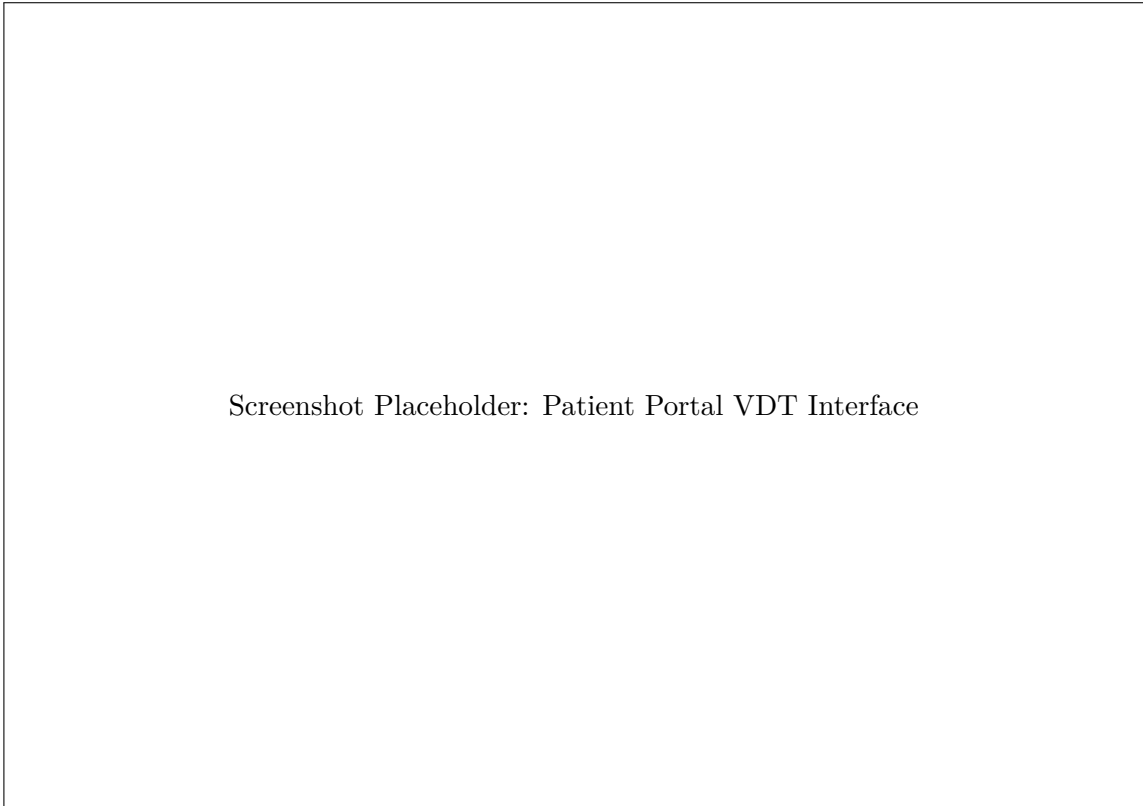
Scenario: Transmit to third party unencrypted with patient consent

Given I want to share my health information
And I accept the risks of unencrypted transmission
When I choose unencrypted transmission
Then the system shall warn me of risks
And system shall transmit unencrypted only with my consent
And my choice shall be documented

Scenario: Access activity history log

Given I have viewed, downloaded, or transmitted my data
When I check my activity history
Then the system shall display activity log
And log shall show what data was accessed
And log shall show when data was accessed
And log shall show to whom data was transmitted

11.3 Screenshots



Screenshot Placeholder: Patient Portal VDT Interface

Figure 11.1: View, Download, Transmit - Patient Portal Interface

Chapter 12

§ 170.315(g)(7) - Application Access - Patient Selection

12.1 Criterion Information

12.1.1 Maturity Level

VERIFIED (behavioral tests green)

12.1.2 Regulatory Text

From 45 CFR § 170.315(g)(7):

(7) Application access—patient selection.

The technology must be able to receive a request with sufficient information to uniquely identify a patient and return an ID or other token.

(Full regulatory text abbreviated. See BDD specification for complete requirements.)

12.2 Behavior-Driven Development Specification

Listing 12.1: 170.315(g)(7) - Application Access

```
# /Volumes/SonicMagic/Code/honeycomb-public-release/certification/bdd/170.315-g-7-10-apis
.feature
# §170.315(g)(7)-(g)(10) - Application Programming Interfaces

# REGULATORY TEXT FROM 45 CFR §170.315(g)(7)
#
# (7) Application accesspatient selection. The following technical outcome and conditions
# must be met through the demonstration of an application programming interface (API).
# (i) Functional requirement. The technology must be able to receive a request with
# sufficient information to uniquely identify a patient and return an ID or other token
# that can be used by an application to subsequently execute requests for that patient
# 's data.
# (ii) Documentation.
# (A) The API must include accompanying documentation that contains, at a minimum:

@170.315-g-7-10 @api @fhir @design-performance
Feature: Application Programming Interfaces
  As a health IT developer
  I want to provide standardized APIs
  So that applications can access health information

Background:
  Given the Health IT Module provides APIs
  And API access is required

Scenario: Provide patient selection API (g)(7)
  Given applications need to select patients
  When application requests patient selection
  Then API shall enable patient selection
  And API shall conform to standards
  And API shall be properly documented

Scenario: Provide data category request API (g)(8)
```

Given applications need specific data categories
When application requests data by category
Then API shall provide data category request capability
And API shall return requested data
And API shall conform to USCDI

Scenario: Provide all data request API (g)(9)

Given applications need complete patient data
When application requests all patient data
Then API shall provide all available data
And data shall include all USCDI elements
And API shall conform to standards

Scenario: Provide standardized API for patient and population services (g)(10)

Given applications need patient and population data
When application uses FHIR API
Then API shall conform to §170.215(a) standard
And API shall support SMART App Launch
And API shall support bulk data access
And API shall provide comprehensive functionality

Chapter 13

§ 170.315(g)(9) - Application Access - All Data

13.1 Criterion Information

13.1.1 Maturity Level

VERIFIED (behavioral tests green)

13.1.2 Regulatory Text

From 45 CFR § 170.315(g)(9):

(9) Application access—all data request.

Provide an application programming interface (API) that can return all available data for a specific patient in a single request.

(Full regulatory text abbreviated. See BDD specification for complete requirements.)

13.2 Behavior-Driven Development Specification

Listing 13.1: 170.315(g)(9) - Application Access

```
# /Volumes/SonicMagic/Code/honeycomb-public-release/certification/bdd/170.315-g-7-10-apis
# .feature
# §170.315(g)(7)-(g)(10) - Application Programming Interfaces

# REGULATORY TEXT FROM 45 CFR §170.315(g)(7)
#
# (7) Application accesspatient selection. The following technical outcome and conditions
# must be met through the demonstration of an application programming interface (API).
# (i) Functional requirement. The technology must be able to receive a request with
# sufficient information to uniquely identify a patient and return an ID or other token
# that can be used by an application to subsequently execute requests for that patient
# 's data.
# (ii) Documentation.
# (A) The API must include accompanying documentation that contains, at a minimum:

@170.315-g-7-10 @api @fhir @design-performance
Feature: Application Programming Interfaces
  As a health IT developer
  I want to provide standardized APIs
  So that applications can access health information

Background:
  Given the Health IT Module provides APIs
  And API access is required

Scenario: Provide patient selection API (g)(7)
  Given applications need to select patients
  When application requests patient selection
  Then API shall enable patient selection
  And API shall conform to standards
  And API shall be properly documented

Scenario: Provide data category request API (g)(8)
```

Given applications need specific data categories
When application requests data by category
Then API shall provide data category request capability
And API shall return requested data
And API shall conform to USCDI

Scenario: Provide all data request API (g)(9)

Given applications need complete patient data
When application requests all patient data
Then API shall provide all available data
And data shall include all USCDI elements
And API shall conform to standards

Scenario: Provide standardized API for patient and population services (g)(10)

Given applications need patient and population data
When application uses FHIR API
Then API shall conform to §170.215(a) standard
And API shall support SMART App Launch
And API shall support bulk data access
And API shall provide comprehensive functionality

Chapter 14

§ 170.315(g)(10) - Standardized API

14.1 Criterion Information

14.1.1 Maturity Level

IMPLEMENTED

14.1.2 Regulatory Text

From 45 CFR § 170.315(g)(10):

(10) Standardized API for patient and population services.

The technology must implement a standardized API that conforms to FHIR R4, US Core, SMART App Launch, OpenID Connect, and SMART Backend Services.

(Full regulatory text abbreviated. See BDD specification for complete requirements.)

14.2 Behavior-Driven Development Specification

Listing 14.1: 170.315(g)(10) - Standardized API

```
# /Volumes/SonicMagic/Code/honeycomb-public-release/certification/bdd/170.315-g-7-10-apis
.feature
# §170.315(g)(7)-(g)(10) - Application Programming Interfaces

# REGULATORY TEXT FROM 45 CFR §170.315(g)(7)
#
# (7) Application accesspatient selection. The following technical outcome and conditions
# must be met through the demonstration of an application programming interface (API).
# (i) Functional requirement. The technology must be able to receive a request with
# sufficient information to uniquely identify a patient and return an ID or other token
# that can be used by an application to subsequently execute requests for that patient
# 's data.
# (ii) Documentation.
# (A) The API must include accompanying documentation that contains, at a minimum:

@170.315-g-7-10 @api @fhir @design-performance
Feature: Application Programming Interfaces
  As a health IT developer
  I want to provide standardized APIs
  So that applications can access health information

Background:
  Given the Health IT Module provides APIs
  And API access is required

Scenario: Provide patient selection API (g)(7)
  Given applications need to select patients
  When application requests patient selection
  Then API shall enable patient selection
  And API shall conform to standards
  And API shall be properly documented

Scenario: Provide data category request API (g)(8)
```

Given applications need specific data categories
When application requests data by category
Then API shall provide data category request capability
And API shall return requested data
And API shall conform to USCDI

Scenario: Provide all data request API (g)(9)

Given applications need complete patient data
When application requests all patient data
Then API shall provide all available data
And data shall include all USCDI elements
And API shall conform to standards

Scenario: Provide standardized API for patient and population services (g)(10)

Given applications need patient and population data
When application uses FHIR API
Then API shall conform to §170.215(a) standard
And API shall support SMART App Launch
And API shall support bulk data access
And API shall provide comprehensive functionality

Chapter 15

§ 170.315(c)(1) - Clinical Quality Measures - Record and Export

15.1 Criterion Information

15.1.1 Maturity Level

VERIFIED (behavioral tests green)

15.1.2 Regulatory Text

From 45 CFR § 170.315(c)(1):

(1) **Clinical quality measures — record and export.**

(i) **Record.** For each clinical quality measure (CQM) presented for certification, record all of the data identified in the standard necessary to calculate the measure.

(ii) **Export.** Enable a user to electronically export a data file, for one or more patients, formatted per the QRDA Category I standard (45 CFR 170.205(h)(2)), that includes all of the recorded CQM data.

(Full regulatory text abbreviated. See BDD specification for complete requirements.)

15.1.3 Implementation

The behavioral test verifies the record-calculate-export pipeline: the measures catalog is populated (seeded CMS/PACIO measures); a numerator event is recorded as a measure-observation-profiled FHIR **Observation**; an individual FHIR **MeasureReport** is calculated (fqm-execution / PACIO evaluators) with initial-population, denominator, and numerator groups and persisted; and the captured reports export as a FHIR Bundle, CSV, JSON, and **QRDA Category I** (§ 170.205(h)(2)). The QRDA I export produces a patient-level CDA document with the QRDA Category I template identifiers and Measure, Reporting Parameters, and Patient Data sections, built from the patient's real demographics and the measure's population results.

Scope: the QRDA I document is real and standards-shaped (correct envelope, template identifiers, and sections). Full per-measure QDM data-criteria entry coverage validated by the Cypress tool is the remaining external certification work; QRDA Category III (for (c)(3)) is a follow-on.

15.2 Behavior-Driven Development Specification

Listing 15.1: 170.315(c)(1) - CQM Record and Export

```
# /Volumes/SonicMagic/Code/honeycomb-public-release/certification/bdd/170.315-c-1-cqm-
  record-export.feature
# §170.315(c)(1) - Clinical Quality Measures - Record and Export

# REGULATORY TEXT FROM 45 CFR §170.315(c)(1)
#
# (1) Clinical quality measures record and export
# (i) Record. For each and every CQM for which the technology is presented for
  certification, the technology must be able to record all of the data that would be
  necessary to calculate each CQM. Data required for CQM exclusions or exceptions must
  be codified entries, which may include specific terms as defined by each CQM, or may
  include codified expressions of patient reason, system reason, or medical reason.
# (ii) Export. A user must be able to export a data file at any time the user chooses and
  without subsequent developer assistance to operate:
```

(A) Formatted in accordance with the standard specified in §170.205(h)(2);
 # (B) Ranging from one to multiple patients; and
 # (C) That includes all of the data captured for each and every CQM to which technology was certified under paragraph (c)(1)(i) of this section.

@170.315-c-1 @cqm-record-export @clinical-quality-measures @cqm

Feature: Clinical Quality Measures - Record and Export

As a healthcare provider

I want to record and export clinical quality measure data

So that I can track quality metrics and submit quality reports

Background:

Given I am authenticated as a provider

And the application has the clinics:quality-measures package installed

Scenario: Record all data necessary for CQM calculation

Given I am documenting patient care for a certified CQM

When I record clinical data elements

Then the system shall enable recording of all data necessary to calculate each CQM

And all required data elements for the CQM shall be capturable

And the data shall be stored in structured format

Scenario: Record codified CQM exclusion data

Given I am documenting a CQM exclusion

When I enter the exclusion reason

Then the system shall accept codified entries for exclusions

And codified entries may include specific terms defined by the CQM

And codified entries may include "patient_reason" expression

And codified entries may include "system_reason" expression

And codified entries may include "medical_reason" expression

Scenario: Record codified CQM exception data

Given I am documenting a CQM exception

When I enter the exception reason

Then the system shall accept codified entries for exceptions

And codified entries may include specific terms defined by the CQM

And codified entries may include reason expressions per CQM requirements

Scenario: Export CQM data at user's discretion

Given CQM data has been recorded for patients

When I choose to export CQM data

Then the system shall enable export at any time I choose

And export shall not require subsequent developer assistance

And I shall be able to initiate export independently

Scenario: Export CQM data in QRPP format

Given CQM data has been recorded

When I export the data

Then the export shall be formatted per §170.205(h)(2)

And the format shall be Quality Reporting Document Architecture QRPP

Scenario: Export CQM data for single or multiple patients

Given CQM data exists for multiple patients

When I configure the export

Then the system shall enable export ranging from one to multiple patients
And I shall be able to select specific patients
Or I shall be able to export data for patient populations

Scenario: Export includes all captured CQM data
Given data has been captured for certified CQMs
When I export CQM data
Then the export shall include all data captured for each certified CQM
And no required data elements shall be omitted from export
And the export shall be complete and accurate

15.3 Screenshots

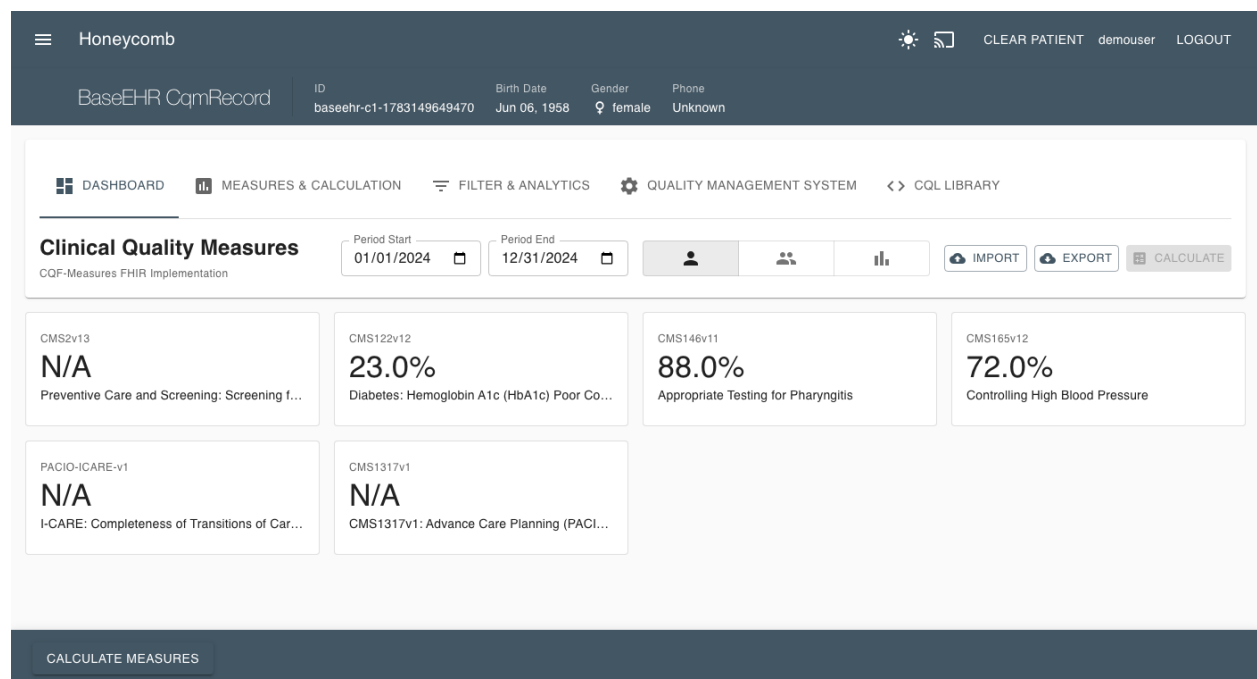


Figure 15.1: Clinical Quality Measures - Record and Export

Chapter 16

§ 170.315(h)(1) - Direct Project

16.1 Criterion Information

16.1.1 Maturity Level

GAP (documented, behavioral tests red)

16.1.2 Regulatory Text

From *45 CFR § 170.315(h)(1)*:

(1) Direct Project.

Send and receive health information using the Direct Standard (Applicability Statement for Secure Health Transport), including SMTP with S/MIME message signing and encryption, message disposition notifications, and trust anchored in X.509 certificates. Criterion (h)(2) additionally requires the Edge Protocols (XDR/XDM) alongside Direct.

(Full regulatory text abbreviated. See BDD specification for complete requirements.)

16.1.3 Documented Gap

The behavioral test verifies the messaging scaffold: the Direct messaging routes render; a Direct address is format-validated (and a malformed address is rejected); and a sent Direct message is persisted as a FHIR **Communication** with the Direct-Message category and an encryption extension, retrievable afterward. These pass green.

Two capabilities are **gaps**, and this is the expected headline Base-EHR gap: (1) there is **no operational Direct transport** — no SMTP+S/MIME send/receive through a Health Information Service Provider (HISP); the configured relay, when present, is plain SMTP, not the Direct Applicability Statement stack (S/MIME signing/encryption, message disposition notifications, DNS CERT / LDAP certificate discovery), and messages persist only as FHIR resources; and (2) **trust verification is simulated** — a placeholder “Demo Certificate Authority” certificate is returned for any well-formed address, with no real X.509 chain validation or DirectTrust bundle. The Edge Protocols (XDR/XDM) required by (h)(2) have no implementation.

Remediation: integrate a real Direct/HISP stack (SMTP + S/MIME per the Applicability Statement, MDN handling, trust-bundle-based certificate validation), and add XDR/XDM for (h)(2). Certification testing uses the ONC Direct Transmission Testing (DTT) tool against a live HISP.

16.2 Behavior-Driven Development Specification

Listing 16.1: 170.315(h)(1) - Direct Project

```
# certification/bdd/170.315-h-1-direct-project.feature
# REGULATORY TEXT FROM 45 CFR §170.315(h)(1)
#
# (1) Direct Project
# (i) Applicability Statement for Secure Health Transport. Able to send and receive
#     health information in accordance with the standard specified in §170.202(a)(2),
#     including formatted only as a wrapped message.
```

(ii) Delivery Notification in Direct. Able to send and receive health information in accordance with the standard specified in §170.202(e)(1).

@170.315-h-1 @direct @transport

Feature: Direct Project

As a healthcare provider

I want to send and receive health information via Direct messaging

So that I can securely exchange patient information with other providers

Background:

Given the system supports Direct Project messaging

And I have appropriate Direct addressing

Scenario: Send health information via Direct with wrapped message

Given I have health information to send

When I send information via Direct messaging

Then the system shall send per §170.202(a)(2)

And the message shall be formatted as a wrapped message

Scenario: Receive health information via Direct with wrapped message

Given health information is being sent to me

When I receive the Direct message

Then the system shall receive per §170.202(a)(2)

And the system shall process the wrapped message

Scenario: Send delivery notification in Direct

Given I have sent health information via Direct

When delivery notification is enabled

Then the system shall send delivery notification per §170.202(e)(1)

Scenario: Receive delivery notification in Direct

Given I sent health information via Direct

When a delivery notification is sent back

Then the system shall receive delivery notification per §170.202(e)(1)

16.3 Screenshots

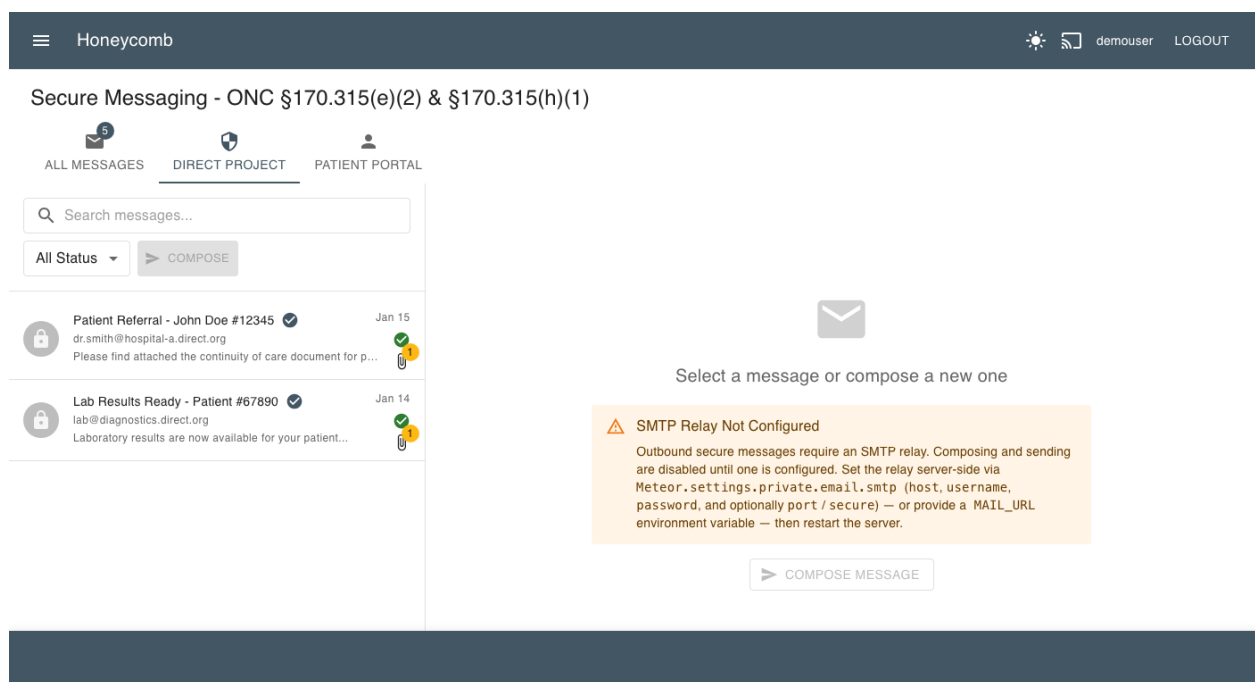


Figure 16.1: Direct Project - Secure Messaging

Appendix A

Acronyms and Abbreviations

API

Application Programming Interface

BDD

Behavior-Driven Development

C-CDA

Consolidated Clinical Document Architecture

CDS

Clinical Decision Support

CFR

Code of Federal Regulations

CPOE

Computerized Provider Order Entry

DI Device Identifier

DSI Decision Support Interventions

EHR

Electronic Health Record

FHIR

Fast Healthcare Interoperability Resources

GMDN

Global Medical Device Nomenclature

GUDID

Global Unique Device Identification Database

HCT/P

Human Cells, Tissues, and Cellular and Tissue-Based Products

HL7

Health Level Seven

MRI

Magnetic Resonance Imaging

OAuth

Open Authorization

ONC

Office of the National Coordinator for Health Information Technology

PI Production Identifier**SMART**

Substitutable Medical Applications, Reusable Technologies

SNOMED CT

Systematized Nomenclature of Medicine - Clinical Terms

SPCU

Sex Parameter for Clinical Use

UDI

Unique Device Identifier

USCDI

United States Core Data for Interoperability

VDT

View, Download, Transmit

Appendix B

References

1. Office of the National Coordinator for Health Information Technology. *2015 Edition Health IT Certification Criteria, 2015 Edition Base EHR Definition, and ONC Health IT Certification Program Modifications*. 45 CFR Part 170. Federal Register, 2015.
2. Office of the National Coordinator for Health Information Technology. *21st Century Cures Act: Interoperability, Information Blocking, and the ONC Health IT Certification Program*. 45 CFR Part 170. Federal Register, 2020.
3. HL7 International. *FHIR R4 Specification*. <http://hl7.org/fhir/R4/>
4. HL7 International. *US Core Implementation Guide*. <http://hl7.org/fhir/us/core/>
5. SMART Health IT. *SMART App Launch Framework*. <http://hl7.org/fhir/smart-app-launch/>
6. Office of the National Coordinator for Health Information Technology. *United States Core Data for Interoperability (USCDI) Version 5*. 2024.
7. FDA. *Unique Device Identification System (UDI System)*. 21 CFR Part 801.
8. FDA. *Global Unique Device Identification Database (GUDID)*. <https://accessgudid.nlm.nih.gov/>

Appendix C

Certification Testing Notes

C.1 Behavioral Test Suite Status

Each Base EHR criterion in Part I has a behavioral Nightwatch test under `certification/tdd/base_ehr/` (one file per criterion, `170.315.<x>.<y>.test.js`). The suite drives real record/change/access behaviors against FHIR resources per the ONC test procedures, not mere page presence, and is run against a live server. As of this revision the outcome is **11 verified** (green) and **1 documented gap** (red): (a)(1), (a)(2), (a)(3), (a)(5), (a)(14), (b)(1), (b)(4), (b)(11), (c)(1), (g)(7), and (g)(9) verify; only (h)(1) Direct Project carries a documented gap. Any remaining gaps use explicit `GAP(170.315.x.y)` annotations that fail deliberately, so the suite functions as a live certification punch-list; the Status column in the Overview and the per-criterion chapters reflect these results. Criterion (g)(10) is covered externally by Inferno (below), not by this suite.

C.2 Inferno Testing

For comprehensive certification testing of the standardized API criterion (g)(10) — and to authoritatively validate the API surfaces exercised by (g)(7) and (g)(9) — the official ONC-approved Inferno testing tool must be used:

Inferno Website: <https://inferno.healthit.gov>

Inferno provides authoritative test suites that validate:

- FHIR R4 conformance
- US Core Implementation Guide compliance
- SMART App Launch OAuth 2.0 flows
- OpenID Connect Core
- SMART Backend Services (Bulk FHIR)
- USCDI data completeness
- Token management and revocation
- All mandatory and must-support elements

The Nightwatch tests in this system provide basic endpoint verification for development and CI/CD purposes. They are **not** a replacement for Inferno certification testing.

C.3 Test Execution

To run the complete Base EHR test suite:

```
cd /path/to/honeycomb-public-release
chmod +x certification/tdd/base_ehr/run-base-ehr-tests.sh
./certification/tdd/base_ehr/run-base-ehr-tests.sh
```

Or run a single criterion directly against a running server:

```
npx nightwatch --config nightwatch.circle.conf.js
  certification/tdd/base_ehr/170.315.<x>.<y>.test.js
```

The suite covers the twelve in-scope CY2026 Base EHR criteria (one test file each); (g)(10) is intentionally excluded and left to Inferno.